

A Unified Treatment of Reachability and Indistinguishability Properties: First-Order Logic with Overwhelming Truth

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Abstract—In the formal verification of complexity-theoretic properties of cryptography, researchers have traditionally attempted to capture “overwhelming truth” (satisfaction with all but negligible probability) via satisfaction on individual traces of probabilistic execution. However, this approach introduces significant complexity when quantification is present: satisfaction of existential quantification over traces often produces witnesses—such as nonce-guessing oracles—that witnesses that, without further constraints, may not correspond to meaningful global objects like PPT algorithms respecting causal structure. This discrepancy creates significant obstacles when attempting to combine trace properties, such as reachability, with properties that hold globally, such as algorithmic indistinguishability.

We resolve this by shifting from local satisfaction on individual traces to a semantics based on ever-decreasing non-negligible sets. We demonstrate that the logical key to this unification lies in first-order modal logic S4 with *non-negligible sets* as possible worlds, rather than the propositional S5 fragment with *traces* as possible worlds suggested in previous investigations by the Squirrel Prover team. By introducing a PPT computational first-order S4 Kripke semantics and adopting Fitting’s embedding for trace properties, we provide a unified quantified treatment of overwhelming truth for trace properties and indistinguishability that admits cut-elimination and remains sound and complete—resolving an open question in the literature.

We show that Fitting’s embedding naturally accommodates the higher-order quantification used in the Squirrel prover by interpreting function types as sorts in a many-sorted first-order logic; this reduces the need for the specialized *const* predicate and its associated structural restrictions. Finally, using our findings, we present a hybrid semantics for CryptoVampire that eliminates the need for bounded Skolemization.

Index Terms—cryptographic protocols, formal verification, computational soundness, modal logic, first-order logic

I. INTRODUCTION

A fundamental challenge in formal verification of cryptographic protocols—when the underlying probabilistic structure along with the security parameter are abstracted out—is combining distinct types of security properties: trace properties that hold with overwhelming probability (all but negligible), properties that require at least non-negligible sets to be meaningful such as PPT computability, and global properties such as computational indistinguishability. Abstraction makes the logic less expressive but simpler and more amenable to automation.

Trace properties concern what happens along individual execution paths—authentication, reachability conditions—while PPT computability only makes sense on non-negligible sets, since on negligible sets such as individual traces, anything can be computed. Indistinguishability, finally, is inherently a relation between entire probability distributions that cannot be localized to individual traces and is not preserved under taking non-negligible subsets. The difficulty lies in providing a unified syntax and semantics where all these types of properties can be reasoned about coherently, especially in the presence of first-order quantification.

To illustrate what can go wrong when we quantify over traces, consider the following from [25]: when existential formulas are evaluated trace-by-trace, witnesses may vary across traces in ways that do not correspond to any meaningful computational object. The classic example is the “nonce-guessing problem”: the formula $\exists x. (n = x)$, where n is an honestly generated nonce and x ranges over some infinite domain independent of n , is satisfied on every trace because some value always equals the nonce on that trace. But this satisfaction does not yield a single PPT algorithm that computes the witness—the Skolem function becomes a non-computable oracle that “guesses” the nonce on each trace.

Furthermore, a unified treatment of trace formulas and indistinguishability is desirable to be able to treat local and global formulas together instead of having to go back and forth between reasoning over traces and over global properties such as indistinguishability. Consider for example the following situation from [2]: Let $[\phi]$ denote that the trace property ϕ holds overwhelmingly. Suppose that we show $[\phi_1] \vdash [\psi]$ and $[\phi_2] \vdash [\psi]$ through a cryptographic argument using indistinguishability. It would be desirable to conclude $[\phi_1 \vee \phi_2] \vdash [\psi]$. However, it is not obvious that we can do that as cryptographic inference on the individual traces is not possible, and of course $[\phi_1 \vee \phi_2]$ does not imply $[\phi_1] \vee [\phi_2]$. Nevertheless, this move is sound if things are defined the right way.

We briefly survey how research groups linked to the Computationally Complete Symbolic Attacker (CCSA) technique (also called BC logic after their creators) have offered varying partial solutions to these issues. The original CCSA technique

for *indistinguishability* [11] avoided local quantification entirely. The domain consists of PPT algorithms, there is only the indistinguishability predicate $\sim$, and satisfaction is considered only over the whole probability space of random tapes. Trace properties are expressed via function symbols $\text{EQ}(_, _)$ and $\text{if } _ \text{ then } _ \text{ else } _$, overwhelming satisfaction is expressed by being indistinguishable from *true*, and with these one can perform propositional logical reasoning over traces and split indistinguishability by cases. The authors in [7] gave a complete axiomatization for such propositional reasoning over traces. While quantification cannot be embedded within trace formulas this way (only prenex quantification is allowed), for a low number of sessions this suffices.

The Squirrel Prover [5], built on the indistinguishability CCSA of Bana and Comon, allows inductive reasoning over the number of sessions, which likely makes quantified reasoning over trace formulas unavoidable. They address this by creating an innovative two-tiered system of combining local and global formulas: second-order quantified local formulas with bit-string domains for trace properties, and multi-sorted first-order quantification for global formulas where the domain consists of *arbitrary random variables*, including those that do not respect the causal structure of the execution. A local trace formula ϕ is not a global formula, but $[\phi]$ meaning that ϕ holds overwhelmingly, is. We cannot grab a PPT algorithm any more with quantification. We cannot say for all nonces, because the meaning of nonce is defined locally, and they do not add up to anything meaningful globally. To avoid the problems this creates, they introduce the $\text{const}(x)$ predicate expressing constancy across traces in a general sense so that you can grab global objects. It is effective but they need associated structural restrictions and machinery for reasoning about them, and a way to combine local and global formulas.

CryptoVampire [25], on the other hand, accepts the satisfiability of $\exists x. (n = x)$ and employs bounded Skolemization to disallow constructing global witnesses from an infinite number of local witnesses. These approaches are all effective but add considerable complexity and provide only partial solutions.

Recently, Antoine and Baelde [2] abstracted the propositional fragment of Squirrel’s treatment of trace formulas with overwhelming satisfaction into reasoning under modal logic S5¹. They presented a modified Kripke semantics where possible worlds are traces, accessibility is the equivalence of all traces and $\Box$ expresses overwhelming satisfaction. They established completeness for the propositional fragment but had to find a workaround for the fact that cut-elimination does not hold in S5. They identify two significant open problems: extending this foundation to first-order quantification, and unifying it with indistinguishability-based reasoning.

In this paper, we resolve both open problems by returning to the original non-Tarskian “twisted” first-order semantics of Bana and Comon [9] (a refinement of [10]) for the CCSA technique introduced for *reachability properties*, and combining it

with their later indistinguishability CCSA, all under first-order S4—not S5. The key insight, developed in [12], is that the original non-Tarskian computational semantics of Bana and Comon is best understood through Fitting’s embedding [23] of classical first-order logic into modal logic S4, combined with a standard Kripke semantics where possible worlds are *non-negligible sets of traces* rather than individual traces, accessibility is the superset relation (S' is accessible from S iff $S' \subseteq S$), and there is a single domain of PPT algorithms. An atomic trace property such as equality is satisfied on a set S if it is satisfied on all traces in S . Fitting proved that if ϕ^* denotes the embedding of a first-order formula ϕ , constructed by placing $\Box\Diamond$ in front of each subformula (we call this the Fitting-twist), then $\Phi \vdash_{\text{FOL}} \phi$ if and only if $\Phi^* \vdash_{\text{FOS4}} \phi^*$, where Φ^* denotes the set of embeddings of formulas in Φ . First-order inference is thus sound and complete for Fitting-twisted formulas, and hence for Bana and Comon’s non-standard semantics.

Note that BC twisted semantics was only a treatment of reachability properties (including PPT computability), and S4 with the Fitting-twist appeared only in the semantics; for the syntax, pure first-order logic was used. On unquantified formulas ϕ , the BC twisted semantics turns out to capture exactly the notion that ϕ is satisfied overwhelmingly: since $\Box\Diamond$ means “for all non-negligible sets there is a smaller non-negligible set,” negligible sets of non-satisfaction can remain. Quantification, on the other hand, ranges over a *global* domain of PPT algorithms (or other appropriate global domains), and the embedding ensures that trace properties of witnesses hold overwhelmingly. Existential quantification in the embedded form $\Box\Diamond\exists x. \phi^*$ means: for every non-negligible set, there is a smaller non-negligible set on which a witness exists, for which ϕ^* holds (that is, ϕ overwhelmingly) on that non-negligible set. Hence the witness is local in the sense that it need only satisfy the required property on an arbitrarily small non-negligible set, but it is an element of the global domain. This eliminates the nonce-guessing problem because on no non-negligible sets can nonces be guessed with non-negligible probability by PPT algorithms. The approach was also generalized to situations requiring satisfaction with sufficient certainty [13], where overwhelming satisfaction is a special case.

Since in this treatment satisfaction of local formulas are defined on non-negligible sets, combining the local and global formulas under an S4 unified treatment does not pose any difficulty. A local formula ϕ is also a global formula, and follows usual Tarski semantics. On the other hand the twisted version $[\phi] \equiv \phi^*$ means that ϕ holds overwhelmingly and it is not the usual Tarski semantics for first-order formulas but nevertheless follows first-order inference rules under $[_]$. Finally, indistinguishability is defined again globally. These all can be unified.

Moreover, S4 provides crucial technical advantages over S5. Unlike S5, S4 admits cut-elimination in standard sequent calculus, enabling cleaner proof-theoretic analysis.

In this paper we focus on 1. extending Antoine and Baelde’s work to quantification through S4, 2. unifying reachability and

¹For very short introduction on modal logic and S4, S5 we refer the reader to Appendix A.

indistinguishability treatment at the global level, 3. simplify Squirrel’s and CryptoVampire’s syntax using a fragment of this technique. To effectively argue with explicit modalities in cryptographic proofs remains future work, though we present some initial results.

Our contributions are as follows:

- **Overwhelming Semantics** We show that for unquantified trace formulas, our overwhelming semantics is exactly the same as the semantics of [2] (and of Squirrel). We also prove that if the global domain is the set of arbitrary random variables, our overwhelming semantics gives exactly Squirrel’s, however, we argue that such global domains should not be used.
- **Soundness and Completeness of FOS4_P**: We prove that the first-order system comprising S4 axioms, constant-domain Barcan formulas, and a persistence axiom $P \rightarrow \Box P$ for atomic trace-property formulas is sound and complete for the combined semantics of overwhelming truth of trace properties with arbitrary global properties. The Barcan formulas are essential for preserving universal properties across refinements of the trace space.
- **Unification of Reachability and Indistinguishability**: Bana and Comon had two separate treatments with first-order logic, one for reachability properties, one for Indistinguishability. We bridge the gap and give a fully unified treatment of reachability-based and indistinguishability-based versions of CCSA under first-order S4. We introduce new inference rules for the combined system.
- **Splitting and patching with Fitting magic**: Unlike the various previous versions of indistinguishability CCSA, the new semantics allows splitting the execution traces: If $[\phi_1] \vdash [\psi]$ and $[\phi_2] \vdash [\psi]$ are provable by some global inference including reasoning about indistinguishability, then a “Fitting magic” combines these together to $[\phi_1 \vee \phi_2] \vdash [\psi]$ even though quantification is not trace-wise.
- **Extension to Higher-Order Features**: We accommodate Squirrel’s multi-sorted syntax by showing that Fitting’s embedding extends naturally to many-sorted first-order logic. By treating function-type sorts as constant domains, we provide a stable bridge between Squirrel’s type system and our modal semantics.
- **Simplification of Squirrel’s Two-Tiered System**: One does not have to go to a full blown first-order S4 inference system. By redefining the interpretation of $[\phi]$ as the BC twisted interpretation of ϕ , and define global domains for types, all of Squirrel’s inference rules retain their soundness, but a number of them can be simplified and new ones can be introduced for splitting and patching. By using Fitting’s embedding with global quantification, we eliminate the need for the $\text{const}(x)$ predicate and its restrictions—“constant-ness” becomes a fundamental property of rigid designators.
- **Hybrid Semantics for CryptoVampire**: We propose a purely first-order twisted hybrid semantics that retains CryptoVampire’s symbolic term algebra while replacing

trace-by-trace evaluation with the BC twisted first-order semantics. This eliminates the nonce-guessing problem semantically, removing the need for bounded Skolemization entirely for free, without introducing new syntax.

In other words, simply by redefining overwhelming satisfaction of trace formulas, that is, changing the semantics, simplifies both Squirrel and CryptoVampire for free, without added syntax, without modalities.

A. Related work.

The landscape of formal verification for cryptographic protocols spans a spectrum from purely symbolic methods to fully computational approaches. At one end, tools like ProVerif [17] operate in the Dolev-Yao symbolic model [22], offering automation but without direct computational soundness guarantees. At the other end, game-based techniques provide full computational security proofs but typically require substantial manual effort.

EasyCrypt [14] and CryptoVerif [18], [20] represent the state of the art in mechanized game-based proofs. EasyCrypt is an interactive theorem prover using probabilistic relational Hoare logic, while CryptoVerif automates proofs via sequences of games formalized in a probabilistic process calculus. CryptHOL [15] provides a framework for game-based proofs in Isabelle/HOL using a semantic domain of generative probabilistic values. SSProve [1] offers a foundational Coq framework for modular game-based proofs using state-separating proof methodology. Recent work has extended CryptoVerif to post-quantum security [19].

The CCSA technique occupies a middle ground: it uses symbolic syntax like Dolev-Yao methods but interprets formulas computationally, axiomatizing cryptographic assumptions directly in first-order logic. This yields computational soundness while retaining much of the simplicity of symbolic reasoning. Both Squirrel [3], [5] and CryptoVampire [25] implement variants of CCSA, with Squirrel supporting both trace properties and indistinguishability, while CryptoVampire focuses on automated reasoning for trace properties via saturation-based theorem provers.

Recent developments in the CCSA ecosystem include PQ-Squirrel [21], which extends the BC logic to post-quantum security by adapting the semantics for quantum attackers, and work by Baelde et al. [4] on SMT-based automation for overwhelming truth reasoning within Squirrel. Bana et al. [8] have developed a Coq formalization of CCSA using Coq’s higher order logic to handle CCSA’s axiom schema and applied it to verify vote privacy in one session of the FOO electronic voting protocol, while more recently, Baelde et al gave an inductive proof in Squirrel for privacy in arbitrary number of sessions or the FOO protocol [6].

II. FIRST-ORDER OVERWHELMING TRUTH

In this section, we first summarize Antoine and Baelde’s propositional logic of overwhelming truth based on S5 [2], then present the alternative first-order semantics based on Fitting’s embedding into S4 [10], [12], [13]. We establish

that the two approaches coincide for quantifier-free formulas, while the S4-based approach extends naturally to first-order quantification.

A. Antoine and Baelde’s Overwhelming Truth in S5

Antoine and Baelde [2] define a modal logic of overwhelming truth where formulas are interpreted over *cryptographic structures*. A cryptographic structure $\mathcal{C}$ consists of an ensemble of sample spaces $\Omega^{\mathcal{C}} = \{\Omega_{\eta}^{\mathcal{C}}\}_{\eta \in \mathbb{N}}$, one for each security parameter η , together with an interpretation of each propositional variable p as an ensemble of functions $p^{\mathcal{C}} = \{p_{\eta}^{\mathcal{C}} : \Omega_{\eta}^{\mathcal{C}} \rightarrow \{0, 1\}\}_{\eta \in \mathbb{N}}$.

a) *Syntax.*: The syntax is standard propositional modal logic:

$$\phi ::= p \mid \perp \mid \phi \Rightarrow \psi \mid \Box \phi$$

where p ranges over propositional variables, and $\Diamond \phi \equiv \neg \Box \neg \phi$. Here the AB subscript stands for Antoine and Baelde.

b) *Semantics.*: The interpretation of each formula ϕ in structure $\mathcal{C}$ is an ensemble of functions $\llbracket \phi \rrbracket^{\mathcal{C}} = \{\llbracket \phi \rrbracket_{\eta}^{\mathcal{C}} : \Omega_{\eta}^{\mathcal{C}} \rightarrow \{0, 1\}\}_{\eta \in \mathbb{N}}$, defined inductively:

- $\llbracket p \rrbracket^{\mathcal{C}} = p^{\mathcal{C}}$
- $\llbracket \perp \rrbracket_{\eta}^{\mathcal{C}}(\rho) = 0$ for all η and $\rho \in \Omega_{\eta}^{\mathcal{C}}$
- $\llbracket \phi \Rightarrow \psi \rrbracket_{\eta}^{\mathcal{C}}(\rho) = 1$ iff $\llbracket \phi \rrbracket_{\eta}^{\mathcal{C}}(\rho) = 0$ or $\llbracket \psi \rrbracket_{\eta}^{\mathcal{C}}(\rho) = 1$
- $\llbracket \Box \phi \rrbracket_{\eta}^{\mathcal{C}}(\rho) = 1$ iff the function $\eta' \mapsto \Pr_{\rho' \in \Omega_{\eta'}^{\mathcal{C}}}[\llbracket \phi \rrbracket_{\eta'}^{\mathcal{C}}(\rho') = 1]$ is *overwhelming*

A function $f : \mathbb{N} \rightarrow [0, 1]$ is *overwhelming* if $1 - f(\eta)$ is negligible, i.e., for all polynomials q , there exists N such that for all $\eta > N$, $1 - f(\eta) < 1/q(\eta)$. Thus $\llbracket \phi \rrbracket^{\mathcal{C}}$ is overwhelming when the function $\eta \mapsto \Pr_{\rho \in \Omega_{\eta}^{\mathcal{C}}}[\llbracket \phi \rrbracket_{\eta}^{\mathcal{C}}(\rho) = 1]$ is overwhelming.

A formula ϕ is *valid* in the logic of overwhelming truth if $\llbracket \phi \rrbracket^{\mathcal{C}}$ is overwhelming for all cryptographic structures $\mathcal{C}$.

c) *Equivalence with S5.*: Antoine and Baelde establish that the modal logic of overwhelming truth coincides exactly with S5. The soundness direction (S5 theorems are valid for overwhelming truth) is relatively straightforward: one verifies that axioms T ($\Box p \Rightarrow p$), 4 ($\Box p \Rightarrow \Box \Box p$), and 5 ($\Diamond p \Rightarrow \Box \Diamond p$) hold under the overwhelming truth interpretation. For completeness, they show that any finite clique counter-model of an S5 formula can be transformed into a cryptographic structure witnessing non-validity.

d) *Determined formulas.*: A key notion in their work is that of *determined formulas*. A formula ϕ is determined if $\Box \phi \vee \Box \neg \phi$ (equivalently, $\Diamond \phi \rightarrow \Box \phi$) is valid—that is, ϕ is either overwhelmingly true or overwhelmingly false, with no middle ground. Determined formulas are those whose Boolean value does not change (overwhelmingly) from trace to trace, and so are constants as discussed in the introduction. Because of the issues mentioned in the introduction they make heavy use of this notion.

e) *Main contribution: hypersequent calculus.*: Since S5 does not admit cut-elimination in standard sequent calculus, Antoine and Baelde’s main technical contribution is adapting Poggiolesi’s hypersequent calculus [27] to handle overwhelming truth.

f) *Limitations.*: Lack of cut-elimination and the need for hypersequent calculus is a significant limitation. Hypersequent calculus significantly complicates the proof system.

Furthermore, Antoine and Baelde explicitly identify two open problems:

- 1) Extending the logic to first-order quantification, which they describe as “very challenging”
- 2) Connecting the logic to the indistinguishability predicate of CCSA

“More generally, it would naturally be desirable to extend our results to richer fragments of the logic of overwhelming truth: one possibility is to extend our language of formulas to the first-order case; it is also possible to stick to the propositional case but incorporate the computational indistinguishability predicate of the CCSA logics. Both directions seem very challenging.”

One cannot incorporate indistinguishability into this semantics: it has to be interpreted at some global layer as indistinguishability cannot be defined on a possible world, that is, a trace. However, there is no global possible world in their semantics.

B. First-Order Overwhelming Truth via Fitting’s Twist

We now present an alternative semantics for overwhelming truth based on Fitting’s embedding of classical first-order logic into modal logic S4 [23]. This approach, developed in [10], [12], [13], works naturally for first-order logic while preserving cut-elimination. The original semantics also contained the computability (derivability) predicate $\triangleright$, which is not a trace property, perhaps giving the impression that the semantics based on Fitting embedding was created only for this particular predicate. However that is not the case. The Fitting embedding works marvelously for any kind of semantics targeting overwhelming truth, that is to say when we need some sort of “enough certainty” as was demonstrated in [13]. For now we focus strictly on trace properties and omit $\triangleright$.

a) *Syntax.*: Bana and Comon build terms from names $n \in \mathcal{N}$ representing honestly generated nonces and function symbols representing PPT computable functions $f \in \mathcal{F}$. We work with standard first-order logic for local formulas. Atomic formulas express trace properties—predicates that may hold or fail on individual random tapes. For protocol verification, typical atomic formulas include:

- Equality: $t_1 = t_2$ (terms t_1 and t_2 evaluate to the same value)
- Other possible trace predicates P

Formulas are built using the standard connectives (for local formulas there is no need for Box and Diamond in the syntax, only in the semantics):

$$\phi ::= P(\vec{t}) \mid \perp \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \rightarrow \psi \mid \neg \phi \mid \forall x. \phi \mid \exists x. \phi$$

b) *Computational model.*: We define a computational model $\mathcal{M}^{\mathcal{C}}$ following the version in [12] with slight modifications. Fix an ensemble of sample spaces $\Omega^{\mathcal{C}} = \{\Omega_{\eta}^{\mathcal{C}}\}_{\eta \in \mathbb{N}}$,

where each Ω_η^C consists of a fixed (independent of η) collection ρ of finite or infinite (at this point it does not matter) random bit strings some representing honest, others representing adversarial randomness.

The computational model $\mathcal{M}^C$ consists of:

- 1) **Domain D :** A set of PPT algorithms operating on the random tapes from Ω^C . We remark that the technique works for other domains or multiple domains corresponding to various types as well such as Quantum computable functions etc. We will see specific examples. The important point is that the domain does not depend on the possible worlds. Each $\alpha \in D$ takes as input the security parameter 1^η and a random tape $\rho \in \Omega_\eta^C$, and outputs a bit string. We write $\alpha_{\eta,\rho}$ for the output of algorithm α on security parameter η and random tape ρ .
- 2) **Variable assignments:** A variable assignment (or valuation) is a function $\nu : \text{Var} \rightarrow D$ mapping each variable to an element of the domain. For $\alpha \in D$, we write $\nu[x \mapsto \alpha]$ for the assignment that maps x to α and agrees with ν on all other variables.
- 3) **Interpretation of function symbols:** Each function symbol f of arity n is interpreted as a PPT computable function $f^{\mathcal{M}^C}$ operating on the random tapes and the *bit-string outputs* of the algorithms in the domain. Given a valuation ν , the interpretation of a term, $\llbracket t \rrbracket^\nu$ at security parameter η and random tape ρ is written $\llbracket t \rrbracket_{\eta,\rho}^\nu$ and defined inductively:
 - For a variable x : $\llbracket x \rrbracket_{\eta,\rho}^\nu = (\nu(x))_{\eta,\rho}$ (the output of the algorithm $\nu(x)$ on η, ρ)
 - For a function application: $\llbracket f(t_1, \dots, t_n) \rrbracket_{\eta,\rho}^\nu = f_\eta^{\mathcal{M}^C}(\llbracket t_1 \rrbracket_{\eta,\rho}^\nu, \dots, \llbracket t_n \rrbracket_{\eta,\rho}^\nu)$
That is, the function is applied to the *outputs* of the subterms on the given trace, not to the algorithms themselves. This ensures locality: the value of a term on a trace depends only on what happens on that trace.
- 4) **Interpretation of predicate symbols:** Each predicate symbol P of arity n representing a trace property (such as equality of output on that trace) is interpreted as a function $P^{\mathcal{M}^C}$ that, given bit-string outputs, returns 0 or 1. For a valuation ν , we write:

$$\llbracket P(\vec{t}) \rrbracket_{\eta,\rho}^\nu = P^{\mathcal{M}^C}(\llbracket t_1 \rrbracket_{\eta,\rho}^\nu, \dots, \llbracket t_n \rrbracket_{\eta,\rho}^\nu)$$

As for functions, to make this a trace property, it can only depend on ρ and the outputs of the interpretations of the arguments. We allow non-PPT computability for generality, but we could just as well have required it.

- 5) **Names:** Each name $n \in \mathcal{N}$ is interpreted as independent uniform random sampling of $p(\eta)$ bits from the honest randomness ρ_h , where p is a fixed polynomial. This is not so important for the time being, it will be important when we come to CryptoVampire.

c) *Kripke semantics over non-negligible sets.*:

Definition 2.1 (Computational Kripke Model): Given a computational model $\mathcal{M}^C$, the associated *computational Kripke*

model is the tuple $\mathcal{M}^{KC} = (W, R, D, \mathcal{I})$ where:

- W is the set of non-negligible subsets of Ω^C (i.e., ensembles $S = \{S_\eta\}_\eta$ where each $S_\eta \subseteq \Omega_\eta^C$ and $\mu(S_\eta)$ is non-negligible as a function of η)
- R is the reverse inclusion relation: SRS' iff $S' \subseteq S$
- D is the domain of PPT algorithms, fixed across all worlds
- $\mathcal{I}$ is the interpretation inherited from $\mathcal{M}^C$: for each function symbol f and predicate symbol P , $\mathcal{I}$ assigns the interpretations $f^{\mathcal{M}^C}$ and $P^{\mathcal{M}^C}$ respectively

The key idea is to interpret formulas in S4 Kripke structures where:

- **Worlds** are non-negligible subsets of Ω^C (more precisely, ensembles $S = \{S_\eta\}_\eta$ where each $S_\eta \subseteq \Omega_\eta^C$ and $\mu(S_\eta)$ is non-negligible as a function of η)
- **Accessibility** is reverse inclusion: SRS' iff $S' \subseteq S$ (equivalently, S is accessible from S')
- **Domain D** is fixed across all worlds (enabling the Barcan formula $\forall x \Box \phi \leftrightarrow \Box \forall x \phi$)

For an atomic formula $P(\vec{t})$ and variable assignment $\nu : \text{Var} \rightarrow D$, satisfaction at world S means $P(\vec{t})$ holds throughout S :

$$\mathcal{M}^{KC}, S, \nu \models_{S4} P(\vec{t}) \iff$$

$$\text{for all } \eta \text{ and all } \rho \in S_\eta, \llbracket P(\vec{t}) \rrbracket_{\eta,\rho}^\nu = 1$$

Logical connectives and quantifiers are interpreted in the standard Tarskian way relative to each world:

- $\mathcal{M}^{KC}, S, \nu \models_{S4} \phi \wedge \psi$ iff $\mathcal{M}^{KC}, S, \nu \models_{S4} \phi$ and $\mathcal{M}^{KC}, S, \nu \models_{S4} \psi$
- $\mathcal{M}^{KC}, S, \nu \models_{S4} \phi \vee \psi$ iff $\mathcal{M}^{KC}, S, \nu \models_{S4} \phi$ or $\mathcal{M}^{KC}, S, \nu \models_{S4} \psi$
- $\mathcal{M}^{KC}, S, \nu \models_{S4} \neg \phi$ iff $\mathcal{M}^{KC}, S, \nu \not\models_{S4} \phi$
- $\mathcal{M}^{KC}, S, \nu \models_{S4} \forall x. \phi$ iff for all $\alpha \in D$, $\mathcal{M}^{KC}, S, \nu[x \mapsto \alpha] \models_{S4} \phi$
- $\mathcal{M}^{KC}, S, \nu \models_{S4} \exists x. \phi$ iff there exists $\alpha \in D$ such that $\mathcal{M}^{KC}, S, \nu[x \mapsto \alpha] \models_{S4} \phi$

Note that modal operators were not included in the syntax for local formulas, but to create an S4 semantics, we need them. This is just a transitional step to recover the non-standard semantics of Bana and Comon. The modal operators capture the structure of non-negligible sets:

- $\mathcal{M}^{KC}, S, \nu \models_{S4} \Box \phi$ iff for all non-negligible $S' \subseteq S$, $\mathcal{M}^{KC}, S', \nu \models_{S4} \phi$
- $\mathcal{M}^{KC}, S, \nu \models_{S4} \Diamond \phi$ iff for some non-negligible $S' \subseteq S$, $\mathcal{M}^{KC}, S', \nu \models_{S4} \phi$

d) *Fitting's embedding.*: Fitting's embedding [23] transforms any first-order formula ϕ into an S4 formula ϕ^* by placing $\Box \Diamond$ in front of every subformula. By S4 equivalences and the Barcan formulas, many of these occurrences can be

simplified, yielding:

$$\begin{aligned}
P(\vec{t})^* &= \Box\Diamond P(\vec{t}) \\
\perp^* &= \perp \\
(\phi_1 \wedge \phi_2)^* &= \phi_1^* \wedge \phi_2^* \\
(\phi_1 \vee \phi_2)^* &= \Box\Diamond(\phi_1^* \vee \phi_2^*) \\
(\phi_1 \rightarrow \phi_2)^* &= \Box(\phi_1^* \rightarrow \phi_2^*) \\
(\neg\phi)^* &= \Box\neg\phi^* \\
(\forall x.\phi)^* &= \forall x.\phi^* \\
(\exists x.\phi)^* &= \Box\Diamond\exists x.\phi^*
\end{aligned}$$

We prefer calling this the Fitting twist. The simplifications follow from S4 properties: $\Box\Diamond$ is idempotent on conjunctions, and the Barcan formula (which holds since the domain is constant across worlds) eliminates $\Box\Diamond$ from universal quantification. For a set of formulas Γ , let Γ^* denote the set of Fitting-twisted formulas.

Fitting's theorem states that $\Gamma \vdash_{\text{FOL}} \phi$ if and only if $\Gamma^* \vdash_{\text{S4}} \phi^*$.

e) BC overwhelming semantics via Fitting.: Bana and Comon used “computational” semantics, did not use the phrase overwhelming semantics. A local formula ϕ holds *computationally* on a non-negligible set S iff its Fitting twist ϕ^* is S4-satisfied on S :

$$\mathcal{M}^C, S, \nu \models_{BC} \phi \iff \mathcal{M}^{KC}, S, \nu \models_{S4} \phi^*$$

We emphasize that using Fitting is a convenience at this point; the final semantics can be defined without modal operators as Bana and Comon did. The *computational satisfaction* relation $\mathcal{M}^C, S, \nu \models_{BC} \phi$ (where S is a non-negligible set and ν is a valuation) is defined recursively as follows:

Definition 2.2 (BC Twisted Computational Semantics): Let $\mathcal{M}^C$ be a computational model, S a non-negligible subset of traces, and $\nu : \text{Var} \rightarrow D$ a valuation into the domain D of PPT algorithms.

Atomic formulas:

- $\mathcal{M}^C, S, \nu \models_{BC} P(\vec{t})$ iff for all non-negligible $S' \subseteq S$, there exists a non-negligible $S'' \subseteq S'$ such that for all η and all $\rho \in S''_{\eta}$, $\llbracket P(\vec{t}) \rrbracket_{\eta, \rho}^{\nu} = 1$. As we will see this is the same as that $1 - \Pr_{\rho}(\rho \in S : \llbracket P(\vec{t}) \rrbracket_{\eta, \rho}^{\nu} = 1)$ is negligible in η .

Propositional connectives:

- $\mathcal{M}^C, S, \nu \models_{BC} \perp$ never holds
- $\mathcal{M}^C, S, \nu \models_{BC} \phi \wedge \psi$ iff $\mathcal{M}^C, S, \nu \models_{BC} \phi$ and $\mathcal{M}^C, S, \nu \models_{BC} \psi$
- $\mathcal{M}^C, S, \nu \models_{BC} \phi \vee \psi$ iff for all non-negligible $S' \subseteq S$, there exists a non-negligible $S'' \subseteq S'$ such that $\mathcal{M}^C, S'', \nu \models_{BC} \phi$ or $\mathcal{M}^C, S'', \nu \models_{BC} \psi$
- $\mathcal{M}^C, S, \nu \models_{BC} \neg\phi$ iff for all non-negligible $S' \subseteq S$, $\mathcal{M}^C, S', \nu \not\models_{BC} \phi$
- $\mathcal{M}^C, S, \nu \models_{BC} \phi \rightarrow \psi$ iff for all non-negligible $S' \subseteq S$, if $\mathcal{M}^C, S', \nu \models_{BC} \phi$ then $\mathcal{M}^C, S', \nu \models_{BC} \psi$

Quantifiers:

- $\mathcal{M}^C, S, \nu \models_{BC} \forall x.\phi$ iff for all $\mathbf{a} \in D$, $\mathcal{M}^C, S, \nu[x \mapsto \mathbf{a}] \models_{BC} \phi$

- $\mathcal{M}^C, S, \nu \models_{BC} \exists x.\phi$ iff for all non-negligible $S' \subseteq S$, there exist a non-negligible $S'' \subseteq S'$ and $\mathbf{a} \in D$ such that $\mathcal{M}^C, S'', \nu[x \mapsto \mathbf{a}] \models_{BC} \phi$

A formula ϕ is *computationally valid* (written $\mathcal{M}^C \models_{BC} \phi$) iff $\mathcal{M}^C, \Omega^C, \nu \models_{BC} \phi$ for all valuations ν (which implies $\mathcal{M}^C, S, \nu \models_{BC} \phi$ for all non-negligible S).

The key insight is that the “for all non-negligible S' , there exists non-negligible S'' ” pattern corresponds exactly to the $\Box\Diamond$ modality in the S4 Kripke semantics, and also ensures overwhelming truth because things are allowed to fail on negligible sets. Only non-negligible sets have to contain others where a formula is satisfied.

Theorem 2.3: If $\Gamma \vdash_{\text{FOL}} \phi$ (i.e., ϕ is derivable from Γ in classical first-order logic), then for any computational model $\mathcal{M}^C$:

$$\mathcal{M}^C \models_{BC} \Gamma \implies \mathcal{M}^C \models_{BC} \phi$$

This soundness theorem is highly non-trivial. Bana and Comon proved it directly in [9], but it follows from Fitting's theorem and S4 soundness for Kripke semantics [12].

Remark 1 (Completeness of first-order inference): The Fitting-based semantics is also complete with respect to the computational semantics. The argument is straightforward: consider any consistent set Γ of first-order formulas. By the completeness of classical first-order logic, Γ has a (standard) first-order model $\mathcal{M}$. This model can be viewed as a degenerate Kripke structure where every non-negligible set is a singleton—the deterministic case where no randomness is involved. In this setting, the Fitting twist becomes trivial ($\Box\Diamond P$ reduces to P for all formulas), and satisfaction in the computational semantics coincides with standard first-order satisfaction.

Remark 2 (Completeness does not mean real attacks): Note that this completeness does not imply that when we find a symbolic attack we can also find a computational one. At this point there are no cryptographic elements, it is just a trivial observation.

C. The BC Semantics and Overwhelming Truth

We now establish that the S4-based Fitting semantics and the S5-based overwhelming truth of Antoine-Baelde [2] coincide for quantifier-free formulas, and extend to first-order formulas over random variable domains.

Proposition 2.4 (Equivalence for Quantifier-Free Formulas): Let ϕ be a quantifier-free formula. Then ϕ holds computationally (i.e., $\mathcal{M}^C \models_{BC} \phi$) if and only if ϕ holds with overwhelming probability in the sense of [2].

Proof:[Proof sketch] By structural induction on ϕ . The base case shows that $\Box\Diamond P$ holds in the Kripke model iff P holds except on a negligible set. The inductive cases for conjunction and negation follow from properties of negligible sets and the Fitting twist. See Appendix B for the full proof. $\square$

The equivalence extends to first-order formulas when the domain consists of all random variables, as in Squirrel's semantics [5].

Definition 2.5 (Random Variable Domain): A random variable domain for a computational model $\mathcal{M}^C$ is a domain D consisting of all measurable functions $\alpha : \Omega^C \rightarrow V$ for some value space V .

Assumption 1 (Measurability of Trace Sets): For any first-order formula ϕ and valuation ν , the set $S_{\phi, \nu} = \{\rho \in \Omega_\eta^C \mid \phi \text{ holds at } \rho \text{ under } \nu\}$ is measurable.

Definition 2.6 (Trace-by-Trace Satisfaction): A formula ϕ holds *trace-by-trace with overwhelming probability* under valuation ν if: $\Pr_{\rho \in \Omega_\eta^C} [\mathcal{M}^C, \{\rho\}, \nu \models \phi] \geq 1 - \text{negl}(\eta)$ where quantifiers range over D (so witnesses can be random variables that depend on ρ).

Theorem 2.7 (Equivalence for Random Variable Domains): Let $\mathcal{M}^C$ be a computational model with random variable domain D , satisfying Assumption 1. For any first-order formula ϕ and valuation ν :

$$\mathcal{M}^C, \Omega^C, \nu \models_{BC} \phi \iff$$

ϕ holds trace-by-trace with overwhelming probability

Proof:[Proof sketch] By structural induction. The key case is existential quantification: when $\exists x. \psi$ holds trace-by-trace, we can construct a witness random variable α^* that picks the correct value at each trace. Since D contains all measurable functions, $\alpha^* \in D$. The converse uses a contradiction argument. See Appendix B for the full proof. $\square$

Remark 3: This means the BC twisted semantics provides exactly the extension of Antoine and Baelde’s S5 semantics to quantification that they seek.

III. COMPLETENESS FOR COMPUTATIONAL MODELS

As long as only twisted semantics is considered (that is, we can put $\Box\Diamond$ everywhere), as was the case in [10] pure first-order logic is sufficient. However, for indistinguishability, this does not work $\Box\Diamond x \sim y$ cannot be used to express the indistinguishability of x and y because indistinguishability is not a local property. Normally x and y are indistinguishable globally while not locally unless they are equal. Consequently, when we want to combine local properties with global ones, we need the full power of S4. We therefore prove a completeness theorem, showing that first-order S4 completely captures computational models and the structure of their non-negligible sets: *The modal logic of computational models together with their non-negligible sets and PPT algorithms as the domain is exactly first-order S4.*

In this section, we establish that first-order S4 logic with the Barcan formulas is sound and complete with respect to our computational semantics. We have seen that for any computational model $\mathcal{M}^C$ and its associated computational Kripke model $\mathcal{M}^{KC} = (W, R, D, \mathcal{I})$ (Definition 2.1), the structure $\mathcal{M}^{KC}$ is an S4 Kripke model with constant domains. The accessibility relation R (reverse inclusion on non-negligible sets) is reflexive and transitive, satisfying the S4 frame conditions. Since the domain D is fixed across all worlds, the Barcan formula and its converse hold. Therefore soundness is immediate since our computational models are

instances of S4 Kripke semantics. The main contribution is completeness: we show that any unprovable formula has a computational counter-model.

A. Propositional Completeness

We define the logic $S4_P$ as standard S4 modal logic augmented with the *Persistence Axiom* for all atomic propositions P expressing trace properties:

$$P \rightarrow \Box P \tag{1}$$

This axiom captures the computational reality that atomic trace properties (such as an equality holding on a specific trace) are stable: if a property holds for an entire set of traces S , it necessarily holds for any subset $S' \subseteq S$.

Definition 3.1 (Finite Rooted Tree Models): A model $\mathcal{M} = (W, R, V)$ is a *finite rooted tree model* if:

- 1) (W, R) forms a finite rooted tree (reflexive, transitive, and anti-symmetric excluding loops).
- 2) The valuation V is persistent: for any $w, u \in W$ and atom P , if wRu and $w \in V(P)$, then $u \in V(P)$.

Theorem 3.2 (Propositional Completeness): A formula ϕ is provable in $S4_P$ if and only if it is valid in all finite rooted tree models.

Proof:[Proof sketch] Soundness is verified by checking that all axioms of $S4_P$ are valid in finite rooted tree models. For completeness, we use standard argument: given an unprovable formula ϕ , we construct a finite counter-model through: (1) building the canonical model for $S4_P$, (2) applying filtration to obtain a finite model, and (3) unraveling into a tree truncated at the modal depth of ϕ . The persistence axiom ensures that the filtration and unraveling preserve the persistence property of valuations. See Appendix B for the full proof. $\square$

To connect this abstract semantics to cryptography, we show that every such tree corresponds to a partition of the underlying probability spaces parametrized by η .

Proposition 3.3 (Computational Realizability): Any finite rooted tree model $\mathcal{M} = (W, R, V)$ can be realized by a computational model $\mathcal{M}^C = (\Omega^C, D, \cdot^{\mathcal{M}^C})$ where worlds correspond to non-negligible sets of traces. Equivalently, there exists a computational Kripke model $\mathcal{M}^{KC}$ (Definition 2.1) whose frame is isomorphic to (W, R) .

Proof:[Proof sketch] We partition the sample space Ω^C among the leaves of the tree, assigning about measure to each leaf. Internal nodes correspond to unions of their descendants’ sets. The accessibility relation is realized by reverse inclusion. See Appendix B for details. $\square$

B. First-Order Completeness

We extend the logic to $FOS4_P$: First-Order S4 with persistent atoms and constant domains. The axioms include all axioms of $S4_P$ (including $P(\vec{t}) \rightarrow \Box P(\vec{t})$ for atomic P), standard first-order axioms and rules, the Barcan formula

$\forall x. \Box \phi \rightarrow \Box \forall x. \phi$, and the converse Barcan formula $\Box \forall x. \phi \rightarrow \forall x. \Box \phi$.

Definition 3.4 (Tree Models with Constant Domains): A tree model $\mathcal{M} = (W, R, D, V)$ consists of a set W of worlds with a distinguished root w_0 , a reflexive transitive relation R where (W, R) forms a tree rooted at w_0 , a fixed (constant) domain D , and a valuation V assigning to each n -ary predicate P and world w a set $V(P, w) \subseteq D^n$, satisfying persistence.

Theorem 3.5 (First-Order Completeness): A formula ϕ is provable in FOS4_P if and only if it is valid in all tree models with constant domains.

Proof:[Proof sketch] Soundness follows from the validity of all axioms in constant domain models. For completeness, we construct the canonical model for FOS4_P and unravel it into a tree. Unlike the propositional case, we do not need filtration or truncation—infinite trees are acceptable since the computational semantics can realize them. See Appendix B for the full proof. $\square$

Proposition 3.6 (Computational Realizability for Tree Models): Any tree model $\mathcal{M} = (W, R, D, V)$ with at most countable branching and at most countable domain can be realized by a computational model.

Proof:[Proof sketch] We assign non-negligible sets to worlds using bits of the random tape: finite branching uses uniform partitioning, while countably infinite branching uses a prefix-free encoding assigning measure $1/2^i$ to the i -th child. The construction is PPT-computable: Witnesses to existential quantifiers are constants while the predicates evaluations depend on which abstract tree corresponds to a particular string on the random tape. See Appendix B for the full proof. $\square$

Corollary 3.7 (Completeness): FOS4_P with constant domains is sound and complete with respect to the class of tree-structured computational models.

IV. ADVANTAGES OF THE S4-BASED APPROACH

The Fitting embedding into S4 offers several advantages over the S5-based approach for reasoning about overwhelming truth. We highlight the key benefits here; further advantages will become apparent when we combine this framework with the indistinguishability-based CCSA technique in Section V.

a) Combining trace properties with indistinguishability. Moving to S4 allows combining overwhelming truth with indistinguishability-based CCSA. Since possible worlds are non-negligible sets (not individual traces), we can define computational indistinguishability on each world using standard definitions. This would be impossible if worlds were individual traces, as indistinguishability is inherently a property of distributions. We can define $[\phi]$ now in the syntax (as opposed to the semantics) as ϕ^* , and $\mathcal{M}^{\text{KC}}, S, \nu \models_{S4} \phi^*$ means overwhelming satisfaction of ϕ on S as before, and first order inference still holds (by Fitting): If $\Gamma \vdash_{\text{FOL}} \phi$ (i.e., ϕ is derivable from Γ in classical first-order logic), then

$$\mathcal{M}^{\text{KC}}, S, \nu \models_{S4} \Gamma^* \implies \mathcal{M}^{\text{KC}}, S, \nu \models_{S4} \phi^*$$

In the *same system* we can also define global predicates such as indistinguishability on any non-negligible set S :

$$\mathcal{M}^{\text{KC}}, S, \nu \models_{S4} \vec{t} \sim \vec{u}$$

b) First-order extension. The Fitting embedding handles quantifiers uniformly:

- $(\forall x. \phi)^* = \forall x. \phi^*$ — universals need no additional modality
- $(\exists x. \phi)^* = \Box \Diamond \exists x. \phi^*$ — existentials receive the standard $\Box \Diamond$ treatment

Both quantification deserves attention.

The existential case deserves careful attention. The formula $\Box \Diamond \exists x. \phi^*$ means: for any non-negligible set S' , there exists a non-negligible subset $S'' \subseteq S'$ and a witness $d \in D$ such that $\phi^*[d/x]$ holds throughout S'' . The witness is *local* in the sense that it need only work on the smaller set S'' , not on all of S' . However, S'' must still be non-negligible—the witness cannot be defined only on a negligible set.

This locality of witnesses is precisely what resolves the Skolemization problem that arises in trace-by-trace semantics. In trace-by-trace approaches (like CryptoVampire’s original semantics), different random strings on the tapes may require different witnesses, and when combined, these witnesses form a non-computable function. In the Fitting semantics, each witness has to work on a non-negligible set, ensuring that Skolem functions remain meaningful and computable.

c) Rigid designators. In our framework, rigid designators (terms that denote the same object across all worlds) arise naturally from the constant domain assumption. There is no need for a separate “constant predicate”; if special global constants are needed, they can be introduced via appropriate typing.

d) Splitting and patching with Fitting magic. Returning to the Introduction’s disjunction problem: suppose we have a proof system like Squirrel’s but in S4. For trace properties ϕ_1, ϕ_2 and ψ , suppose cryptographic arguments using indistinguishability yield with global Γ all boxed formulas:

$$\Gamma, [\phi_1] \vdash [\psi] \quad \text{and} \quad \Gamma, [\phi_2] \vdash [\psi]$$

In our S4 setting this corresponds to $\Gamma, \phi_1^* \vdash \psi^*$ and $\Gamma, \phi_2^* \vdash \psi^*$ so $\Gamma, \phi_1^* \vee \phi_2^* \vdash \psi^*$. In S4, if the formulas in Γ all have boxes outside, we have $\Gamma, \Box \Diamond (\phi_1^* \vee \phi_2^*) \vdash \Box \Diamond \psi^*$. Since ψ^* already has $\Box \Diamond$ outside, $\Box \Diamond \psi^* \leftrightarrow \psi^*$, while $\Box \Diamond (\phi_1^* \vee \phi_2^*) \equiv (\phi_1 \vee \phi_2)^*$. Therefore $\Gamma, (\phi_1 \vee \phi_2)^* \vdash \psi^*$, i.e.,

$$\Gamma, [\phi_1 \vee \phi_2] \vdash [\psi].$$

One also obtains $[\phi_1 \vee \phi_2 \rightarrow \psi]$: from $(\phi_1 \vee \phi_2)^* \vdash \psi^*$ we have $\vdash (\phi_1 \vee \phi_2)^* \rightarrow \psi^*$. By necessitation, $\vdash \Box ((\phi_1 \vee \phi_2)^* \rightarrow \psi^*)$, which is $\vdash (\phi_1 \vee \phi_2 \rightarrow \psi)^*$. The magic works because since the twisted semantics has a box outside, $[\phi] \vdash [\psi]$ means not only that if $[\phi]$ is true on the whole space then $[\psi]$ is also true, but that whenever $[\phi]$ holds on a non-negligible set, $[\psi]$ must also hold. Such situations arise when protocols branch—some keys generated on one side, others on another, with the same authentication result but for different reasons. One can

do the proofs separately, but cryptographic assumptions have to be written in the form $\Box(\Phi \rightarrow x \sim y)$ to make sure that ϕ contains all the conditions that we are in a part of the execution where all key generation and randomness for primitives' security are contained. This remains future work.

e) **Determined formulas.:** Determined formulas, if still needed, can also be treated in this framework: They cannot be defined with an embedded formula (just as in AB's case), we have to go out to S4: A formula ϕ is *determined* if $\phi^* \vee (\neg\phi)^*$ is valid—it is either overwhelmingly true or overwhelmingly false. In the S4 Kripke setting over non-negligible sets, determined formulas correspond to those whose truth value is constant across all worlds.

f) **Cut-elimination.:** S4 admits cut-elimination in standard sequent calculus, whereas S5 requires hypersequent calculus. This yields simpler proof theory and more direct metatheoretic results. For protocol verification, this means proofs can be constructed and verified using standard techniques. The addition of $P \rightarrow \Box P$ does complicate the picture, but cut-elimination does hold if one moves from a standard calculus to a labeled sequent calculus [26].

V. CCSA TECHNIQUE BASED ON INDISTINGUISHABILITY

The Computationally Complete Symbolic Attacker (CCSA) technique based on indistinguishability was introduced by Bana and Comon-Lundh in [11] as a follow-up to their original work on reachability properties [10]. This version of the CCSA technique represented the real breakthrough of the approach, as it provides a computationally sound symbolic technique that is syntactically simple, and standard game-based computational security notions are easy to translate to it. We summarize the key elements following [7]; the interested reader should consult that paper for complete details, including full axiom formulations and soundness proofs. Both the Squirrel prover [3], [5] and CryptoVampire [25] are based on this indistinguishability version of CCSA, though with different design choices regarding quantification and the treatment of trace properties.

A. Syntax and Semantics

The syntax is based on first-order logic with a distinguished predicate symbol $\sim$ representing computational indistinguishability. Terms are built from a finite set of sorts $\mathcal{S}$ including at least **bool** and **message**. The set of names $\mathcal{N}$ (of type **message**) serves as function symbols of arity zero (constants) representing random seeds, interpreted as uniformly random bit strings. Function symbols $\mathcal{F}$ include basic operations like **0** (empty message), **true**, **false**, $\text{EQ}(_, _)$ for equality testing, and **if _ then _ else _** for conditional branching. Functions symbols for cryptographic operations representing encryption, decryption, signature, and possibly other deterministic functions are also represented by symbols in $\mathcal{F}$. Additionally, a countable set $\mathcal{G}$ represents adversarial function symbols.

The key atomic formula is $t_1, \dots, t_n \sim u_1, \dots, u_n$, which represents that the two sequences of terms are computationally indistinguishable. Formulas are standard first-order formulas

built on these atomic formulas. The abbreviation $x = y$ stands for $\text{EQ}(x, y) \sim \text{true}$, that x and y are equal with overwhelming probability.

a) **Global formulas.:** The syntax of global formulas is defined inductively:

- **Indistinguishability atoms:** $t_1, \dots, t_n \sim u_1, \dots, u_n$ where t_i, u_i are terms
- **Boolean connectives:** $\neg\phi, \phi \wedge \psi, \phi \vee \psi, \phi \rightarrow \psi$
- **Quantifiers:** $\forall x.\phi$ and $\exists x.\phi$ where x is a variable of any sort

Recall that computational equality $t = u$ is the abbreviation $\text{EQ}(t, u) \sim \text{true}$, meaning that $\Pr_\rho[\llbracket t \rrbracket_{\eta, \rho} = \llbracket u \rrbracket_{\eta, \rho}]$ is overwhelming in η .

The computational semantics is a *standard Tarski semantics* that interprets terms as probabilistic polynomial-time algorithms. A computational model $\mathcal{M}^c$ extends the framework of Section II-B with a key distinction: function symbols are partitioned into *honest* symbols $\mathcal{F}$ (interpreted as deterministic algorithms using only their inputs) and *adversarial* symbols $\mathcal{G}$ (interpreted as PPT algorithms with access to adversarial randomness). Nevertheless, that is not too relevant here. Importantly:

- The equality test $\text{EQ}(t, u)$ is interpreted as returning **true** if $\llbracket t \rrbracket_{\eta, \rho}^\nu = \llbracket u \rrbracket_{\eta, \rho}^\nu$ and **false** otherwise:

$$\llbracket \text{EQ}(t, u) \rrbracket_{\eta, \rho}^\nu = \begin{cases} \text{true} & \text{if } \llbracket t \rrbracket_{\eta, \rho}^\nu = \llbracket u \rrbracket_{\eta, \rho}^\nu \\ \text{false} & \text{otherwise} \end{cases}$$

- The conditional **if b then u else v** is interpreted as:

$$\llbracket \text{if } b \text{ then } u \text{ else } v \rrbracket_{\eta, \rho}^\nu = \begin{cases} \llbracket u \rrbracket_{\eta, \rho}^\nu & \text{if } \llbracket b \rrbracket_{\eta, \rho}^\nu = \text{true} \\ \llbracket v \rrbracket_{\eta, \rho}^\nu & \text{otherwise} \end{cases}$$

Given a variable assignment (valuation) ν mapping variables to elements of D (as in Section II-B), a term t is interpreted as $\llbracket t \rrbracket_{\eta, \rho}^\nu \in D$, or if it is a boolean then $\llbracket t \rrbracket_{\eta, \rho}^\nu \in D_b$. The indistinguishability predicate is interpreted as: $\mathcal{M}^c, \nu \models t_1, \dots, t_n \sim u_1, \dots, u_n$ if and only if for any polynomial-time distinguisher $\mathcal{A}$,

$$\left| \Pr_\rho[\mathcal{A}(\llbracket \vec{t} \rrbracket_{\eta, \rho}^\nu; \rho_2) = 1] - \Pr_\rho[\mathcal{A}(\llbracket \vec{u} \rrbracket_{\eta, \rho}^\nu; \rho_2) = 1] \right|$$

is negligible in η . Here $\llbracket \vec{t} \rrbracket_{\eta, \rho}^\nu$ is the output of $\llbracket t \rrbracket_{\eta, \rho}^\nu$ on input security parameter 1^η and fixed randomness ρ . Contrary to the reachability based CCSA, in indistinguishability CCSA connectives and quantifiers are interpreted the standard Tarski way.

B. Axiomatization

The technique employs a library of axioms that fall into two categories. *Core axioms* are independent from protocols and primitives, formalizing basic properties of indistinguishability, random generation, and the conditional construct. These include axioms like:

- **REFL:** $t \sim t$
- **SYM:** $\vec{t} \sim \vec{u} \longrightarrow \vec{u} \sim \vec{t}$
- **TRANS:** $\vec{t} \sim \vec{u} \wedge \vec{u} \sim \vec{v} \longrightarrow \vec{t} \sim \vec{v}$

- **FUNCAPP**: $\vec{t} \sim \vec{u} \longrightarrow f(\vec{t}) \sim f(\vec{u})$
- **IFTRUE**: $b \sim \text{true} \longrightarrow \text{if } b \text{ then } u \text{ else } v \sim u$
- **IFFALSE**: $b \sim \text{false} \longrightarrow \text{if } b \text{ then } u \text{ else } v \sim v$

Cryptographic axioms formalize hardness assumptions and security properties of primitives. The axioms from [7] are:

- **DDH** (Decisional Diffie-Hellman) axiom
- **IND-CPA**, **IND-CCA1**, **IND-CCA2** axioms for encryption
- **UF-CMA** (Unforgeability under Chosen Message Attack) axiom for signatures

These axioms are formalized in [7] such that recursively defined constraints on contexts represent execution traces of the games corresponding to the security notion, and the indistinguishability of these execution traces are stated. A complete axiomatization for `if _ then _ else _` with booleans was also given.

a) *Example: IND-CPA and IND-CCA axioms.*: To illustrate the axiom formulation, consider the encryption security axioms. Let $k(\cdot)$ generate public-secret key pairs, with $\text{pk}(n) \equiv \pi_1(k(n))$ and $\text{sk}(n) \equiv \pi_2(k(n))$. Let $\{m\}_{\text{pk}(n)}^r$ denote encryption of message m with public key $\text{pk}(n)$ using randomness name r . Let $L(\cdot)$ return the length of a message. A formula $\vec{t}[x] \sim \vec{u}[x]$ is called:

- **ENC_{CPA}** if $\text{sk}(n)$ does not occur anywhere in $\vec{t}, \vec{u}$
- **ENC_{CCA1}** if $\text{sk}(n)$ occurs only in decryption positions $\text{dec}(t', \text{sk}(n))$ where t' does not contain x
- **ENC_{CCA2}** if $\text{sk}(n)$ occurs only in decryption positions guarded by `if EQ( $t', x$ )  $\wedge$  EQ( $L(m), L(m')$ ) then ... else  $\text{dec}(t', \text{sk}(n))$`

Here x represents the “challenge ciphertext” position. The axiom schema then states that if $\vec{t}[x] \sim \vec{u}[x]$ satisfies the appropriate constraint, and n_2, n_3 are fresh names not occurring elsewhere, then:

$$\begin{aligned} & \vec{t} \left[\text{if EQ}(L(m), L(m')) \text{ then } \{m\}_{\text{pk}(n)}^{n_2} \text{ else } u'' \right] \\ & \sim \vec{u} \left[\text{if EQ}(L(m), L(m')) \text{ then } \{m'\}_{\text{pk}(n)}^{n_3} \text{ else } u'' \right] \end{aligned}$$

The length check $\text{EQ}(L(m), L(m'))$ ensures the plaintexts have equal length (as required by the IND-CPA/CCA definitions). The constraints on where $\text{sk}(n)$ may appear directly encode the game restrictions: CPA allows no decryption oracle access, CCA1 allows decryption only before receiving the challenge, and CCA2 allows decryption even after, except on the challenge ciphertext itself. Strict recursive definitions were given in [7].

In CCSA it is not indistinguishability of two games that are proven as with game reduction proofs, but instead indistinguishability of two executions of the games. A sequence of attacker computations interacting with the game are represented as just function symbols f_1, f_2 , etc. that are the same on the two sides and that act on the previous output of the challenger (or the protocol). In other words, say we have two protocols or games Π_l and Π_r the equivalence of which expresses the

security property. Then these are unfolded into a sequence of branching terms

$$\Phi(\Pi_{l/r})_{i+1} = g_{l/r}(f_{i+1}(\Phi(\Pi_l)_i)).$$

Here $f_{i+1}(\Phi(\Pi_{l/r})_i)$ represents the attacker’s computation in the current round. Contrary to Dolev-Yao, this is not a function symbol representing some concrete computation like decryption, but an abstract symbol representing any PPT computation. On the other hand, $g_{l/r}$ represents the challenger or agent honest branching computation using `if  $b$  then  $u$  else  $v$` and **EQ** to do a test b on the incoming attacker message $f_{i+1}(\Phi(\Pi_{l/r})_i)$ and output u or v depending on the test result where these terms are computed from previous messages. Note that the attacker computations are the same on both sides, they do not depend on l/r . The aim of the security proof is that using the aforementioned axioms show that the sequence of terms obtained this way are equivalent $\Phi(\Pi_l) \sim \Phi(\Pi_r)$. The branching can be taken apart in the course of the proof using branching axioms like $(b, u \sim b', u') \rightarrow (b, t \sim b', t') \rightarrow (b, \text{if } b \text{ then } u \text{ else } t \sim b', \text{if } b' \text{ then } u' \text{ else } t')$, but indistinguishability is proven across the whole execution.

The main soundness theorem [11] states: Assuming that only bounded (but with arbitrary bound) number of session calls are allowed by any PPT attacker, if a set of axioms A (representing cryptographic assumptions) together with the negation of an indistinguishability statement for two protocols are inconsistent (i.e., $A \cup \{\Phi(\Pi_l) \not\sim \Phi(\Pi_r)\}$ has no first-order model), then the protocol Π_l and Π_r are computationally indistinguishable under those assumptions.

VI. COMBINING INDISTINGUISHABILITY AND OVERWHELMING TRUTH OF TRACE PROPERTIES

In this section we start linking the reachability CCSA’s semantics through Fitting’s embedding, and the indistinguishability CCSA’s Tarskian first-order semantics. While the indistinguishability framework is primarily designed for equivalence properties, trace properties can also be expressed. In fact often a large part of the proofs are arguments about trace properties. A trace property “ ϕ holds with overwhelming probability” can be encoded as $b_\phi \sim \text{true}$ where b_ϕ is a Boolean term corresponding to ϕ . The if-then-else construct `if  $b$  then  $u$  else  $v$` plays a crucial role: it can represent both protocol branching and case analysis in proofs. As we will see, this provides a connection between the indistinguishability framework and the overwhelming truth semantics of trace formulas via Fitting embedding.

A. If-Then-Else in the Indistinguishability Framework

The key to connecting the two CCSA approaches lies in the if-then-else construct. In the indistinguishability formulation, trace properties are formalized using terms of the form `if  $b$  then  $u$  else  $v$` , where `if _ then _ else _` is a function symbol that can express either:

- An honest agent’s execution step based on checking some Boolean condition b
- A case distinction introduced in the course of the proof

Combining such terms, one can express Boolean operations. For example:

- if b then (if b' then u else v) else v is equivalent to if $b \wedge b'$ then u else v
- if b then u else v is equivalent to if $\neg b$ then v else u

When u and v are also terms of type `bool`, then we effectively obtain a propositional calculus over traces. As is shown in [7], it is easy to provide a complete axiomatization of this propositional calculus with the `if then else` function symbol. It was also shown in the same publication that for any b of type `bool`, we have

$$b \sim \text{true} \longleftrightarrow b = \text{true}.$$

B. Translation Between If-Then-Else and Local Formulas

Let us look back at the local formulas of Section II-B. Any local predicate P can be viewed as a function symbol of type `bool`. The function symbol corresponding to equality is `EQ`, but for other predicates we will just use the same notation.

To any local formula ϕ we assign a Boolean term b_ϕ :

- $b_{P(\vec{t})} \equiv P(\vec{t})$
- $b_{\neg\phi} \equiv \text{if } b_\phi \text{ then false else true}$
- $b_{\phi \wedge \psi} \equiv \text{if } b_\phi \text{ then } b_\psi \text{ else false}$
- $b_{\phi \vee \psi} \equiv \text{if } b_\phi \text{ then true else } b_\psi$
- $b_{\phi \rightarrow \psi} \equiv \text{if } b_\phi \text{ then } b_\psi \text{ else true}$

The inverse is also possible of course, but this way is easier to describe.

It is immediate from the definitions that for any local formula ϕ without quantifiers,

$$\phi^* \leftrightarrow [\phi] \leftrightarrow b_\phi = \text{true}$$

as each of these mean overwhelming truth. In other words, a local formula ϕ without quantifiers is overwhelmingly satisfied if and only if b_ϕ evaluates to `true` overwhelmingly. The same can be done when we quantify over arbitrary random variables as explained later.

This has important consequences. It means that any Boolean term can be directly translated to an unquantified (for us) formula, and the twisted semantics of this formula corresponds exactly to the Boolean term being indistinguishable from (that is overwhelmingly equal to) `true`.

C. Unification in S4 Through Fitting's Embedding

At this point it is entirely clear how to unify the two methods. They are unified under the S4 semantics where local quantified formulas get translated by Fitting, while formulas containing the equivalence predicate stay the same. The domain of interpretation for both of them is still D regardless whether we consider local or global formulas. We only have to define the semantics of $\vec{u} \sim \vec{v}$ over arbitrary non-negligible set, which is easy: $\mathcal{M}^{\text{CC}}, S, \nu \models \vec{t} \sim \vec{u}$ iff

$$\left| \Pr_\rho[\mathcal{A}(\llbracket \vec{t} \rrbracket_{\eta, \rho}^\nu; \rho_2) = 1 \mid \rho \in S] - \Pr_\rho[\mathcal{A}(\llbracket \vec{u} \rrbracket_{\eta, \rho}^\nu; \rho_2) = 1 \mid \rho \in S] \right|$$

is negligible.

a) *Unified syntax.*: The unified logic combines local formulas (trace properties) with global formulas (indistinguishability statements) under a single S4 framework. Any local formula is also a global formula. For a purely local formula ϕ , the notation $[\phi]$ is an abbreviation for ϕ^* .

Global formulas Φ are S4 formulas built from:

- Atomic: local predicates $P(\vec{t})$ (including equality), and indistinguishability $\vec{t} \sim \vec{u}$
- Compound: $\neg\Phi, \Phi \wedge \Psi, \Phi \vee \Psi, \Phi \rightarrow \Psi$
- Possibly other global predicates
- Quantified: $\forall x. \Phi, \exists x. \Phi$
- Modal: $\Box\Phi, \Diamond\Phi$

Semantics: Given computational model $\mathcal{M}^C$ and its associated computational Kripke model $\mathcal{M}^{\text{CC}}$ (Definition 2.1), non-negligible set S , and valuation ν :

- $\mathcal{M}^{\text{CC}}, S, \nu \models P(\vec{t})$ iff $P(\vec{t})$ holds for all traces in S
- $\mathcal{M}^{\text{CC}}, S, \nu \models \vec{t} \sim \vec{u}$ iff for all PPT $\mathcal{A}$:

$$\begin{aligned} & \left| \Pr[\rho \in S : \mathcal{A}(\llbracket \vec{t} \rrbracket_{\eta, \rho}^\nu; \rho_2) = 1] \right. \\ & \left. - \Pr[\rho \in S : \mathcal{A}(\llbracket \vec{u} \rrbracket_{\eta, \rho}^\nu; \rho_2) = 1] \right| \leq \text{negl}(\eta) \end{aligned}$$

Where ϕ_2 is an attacker tape.

- Connectives and quantifiers: standard Tarskian interpretation at world S
- $\mathcal{M}^{\text{CC}}, S, \nu \models \Box\Phi$ iff for all non-negligible $S' \subseteq S$, $\mathcal{M}^{\text{CC}}, S', \nu \models \Phi$
- $\mathcal{M}^{\text{CC}}, S, \nu \models \Diamond\Phi$ iff for some non-negligible $S' \subseteq S$, $\mathcal{M}^{\text{CC}}, S', \nu \models \Phi$

The abbreviation $[\phi]$ for a local formula ϕ is interpreted as ϕ^* , i.e., $\mathcal{M}^{\text{CC}}, S, \nu \models [\phi]$ iff $\mathcal{M}^{\text{CC}}, S, \nu \models \phi^*$.

D. Inference Rules and Soundness

All axioms of reachability CCSA remain sound in the new S4 semantics, because each possible world is a model for those axioms. As for indistinguishability axioms, some of them are sound for any S like the equivalence property of $\sim$ and the axiomatization of `if _ then _ else _`. All of them are sound for the entire execution space $\mathcal{M}^C$ as global properties, but not for arbitrary subsets. To investigate how to define preconditions to cryptographic properties to allow satisfaction on any possible world (in the form $\Phi \rightarrow x \sim y$) remains future work. For example $S \models n \sim n'$ can only hold for two nonces if S is such that on it n and n' are honestly and *completely* generated. We need Φ to ensure this. In other words, currently at the *global level*, we can effectively combine the two systems at the global level using the semantics of ϕ^* and $x \sim y$ for indistinguishability—as we show for Squirrel—but we need more work to effectively argue with explicit modalities. Nevertheless, here are some sound inference rules:

- First-order S4 inference with Barcan formulas
- For atomic trace formulas $P(\vec{t}) \rightarrow \Box P(\vec{t})$
- For ϕ without quantification, $\phi^* \leftrightarrow b_\phi = \text{true}$ as explained before.
- $\Box\Diamond\vec{x} \sim \vec{y} \rightarrow x = y$. This axiom is sound because if $\Box\Diamond\vec{x} \sim \vec{y}$ holds but there is a non-negligible set S on which $x \neq y$, then there must be a further non-negligible

set S' on which the interpretations of x and y are set apart by some inequality. This inequality test can differentiate the two interpretations on S' contradicting the premise.

- Let ϕ be either a formula without quantifiers or quantified only over the set of arbitrary random variables. Then according to Proposition 2.4 and Theorem 2.7, we can identify S_ϕ where ϕ holds and $S_{\neg\phi}$ where it does not. Let Γ contain only formulas that have $\Box$ outside like the Fitting-twisted ones.

$$\frac{\Gamma, \phi^* \vdash \vec{x} \sim \vec{y}, \quad \Gamma, (\neg\phi)^* \vdash \vec{x} \sim \vec{y}}{\Gamma \vdash \vec{x} \sim \vec{y}}$$

This is sound since $S_{\phi^*} \cup S_{(\neg\phi)^*} = \Omega^C$ (modulo the negligible boundary $\partial\phi$), the probability space is partitioned into two stable ensembles. Because $u \sim v$ holds in both partitions and the boundary is negligible, the advantage of any PPT distinguisher on the whole space is the sum of negligible advantages, which is itself negligible. It is necessary that the formulas in Γ also hold on S_{ϕ^*} and $S_{(\neg\phi)^*}$, that is why the $\Box$ is required outside.²

Contrary to the branching axiom mentioned at the end of the previous section, this axiom allows the one to actually split the execution space into parts, one on which ϕ holds and another on which $\neg\phi$ and prove the equivalences on the branches. The axiom has to be applied with care though, as splitting with the wrong condition may result two branches on none of which indistinguishability holds while on the whole space it might. But this is also true for the previous branching axiom.

VII. GLOBAL AND LOCAL FORMULAS IN SQUIRREL PROVER

The Squirrel prover [3] initially defined a meta-logic on the top of the CCSA technique, however subsequently refined their technique in [5] moved much closer to how [7] treat CCSA, in particular how axioms expressing the security of primitives are formalized using recursive functions as the security game execution such as the case of IND-CPA, IND-CCA.

Nevertheless, there are crucial differences, some we hinted earlier. One crucial difference is that Squirrel defines satisfaction of local formulas over traces, causing the aforementioned host of issues and forcing them to allow a global domain of arbitrary random variables, including those that are not PPT and those that do not even observe the causal structure of a protocols execution—in other words, depend on the future.

Another important difference is that Squirrel now allows quantification of term contexts, that is, quantification over functions. For example, one can write $\forall f : \text{message} \rightarrow \text{message}. \phi(f)$ where f ranges over functions from messages to messages. In the simply-typed λ -calculus notation used by Squirrel, such functions are represented as $\lambda x.t$ where t is a term possibly containing x . The key point is that on each trace ρ , such a function f determines a concrete function from bitstrings to bitstrings, so quantification over f at the

local level means: for every choice of such a function on the current trace. When the target type of the quantified f is `bool`, then locally over traces this means quantification over local predicates making local formulas second-order over traces.

A further crucial difference is the following: although Theorem 5.2 in [7] implies that any propositional tautology expressed with `if then else` on a trace can be derived using a few axioms for `if _ then _ else _`, for Squirrel’s inductive reasoning on trace properties it is more convenient to treat local properties as separate “local formulas” rather than under `if _ then _ else _`. But since they go back and forth between the local formulas and `if _ then _ else _`, they also introduced boolean quantification function symbols.

A. The Two-Tiered Structure. Syntax and Semantics

The key distinction in Squirrel is between two levels of formulas with fundamentally different quantification domains.

a) Local formulas: In Squirrel, local formulas are essentially syntactic sugar for terms of type `bool`. The logic introduces quantification function symbols $\forall_\tau$ and $\exists_\tau$ for each type τ , both with type $(\tau \rightarrow \text{bool}) \rightarrow \text{bool}$, interpreted with the expected semantics. Propositional connectives are defined using `if-then-else`: for instance, $\phi \wedge \psi$ stands for `if  $\phi$  then  $\psi$  else false` as we have seen earlier. Thus local formulas are terms of type `bool`, representing Boolean-valued random variables. They are interpreted trace-by-trace: a local formula ϕ evaluates to *true* or *false* depending on the random tape ρ and security parameter η . Local formulas include:

- *Equations:* $t_1 = t_2$ (as Boolean terms)
- *Connectives:* Built using `if-then-else`: $\phi \wedge \psi \equiv \text{if } \phi \text{ then } \psi \text{ else false}$
- *Local quantifiers:* $\forall x.\phi$ and $\exists x.\phi$ where x ranges over values on the current trace. These are also defined using the quantification function symbols: $\forall x.\phi$ is syntactic sugar for $\forall_\tau(\lambda x.\phi)$ (where of course ϕ is also syntactic sugar in the above sense. The semantics of `if _ then _ else _` is the same as before, while for universal quantification:

$$\llbracket \forall_\tau(\lambda x.\phi) \rrbracket_{\eta, \rho} = 1 \text{ iff } \llbracket \phi \rrbracket_{\eta, \rho}^{[x \mapsto a]} = 1 \text{ for all } a \in \llbracket \tau \rrbracket_\eta$$

Crucially, local quantification is trace-bound: when evaluating $\forall x.\phi$ on trace ρ , the variable x ranges over values available on that specific trace. This includes quantification over function symbols of type $\tau_1 \rightarrow \tau_2$, where the function’s meaning is determined trace-by-trace. Since one can quantify over function symbols of type `bool`, local formulas effectively have second-order expressive power over individual traces.

b) Global formulas: are first-order formulas interpreted over the domain of *random variables*—functions from traces to values. These include:

- *Overwhelming truth atoms:* $[\phi]$ asserts that local formula ϕ holds with overwhelming probability
- *Indistinguishability predicate:* $\tilde{t}_1 \sim \tilde{t}_2$ asserts computational indistinguishability
- *Other predicates:* $\text{const}(t)$ (term is constant across traces), $\text{adv}(t)$ (term is adversarially computable)
- *First-order connectives:* $\tilde{\wedge}, \tilde{\vee}, \tilde{\exists}, \tilde{\exists}$

²Hideki Sakurada had posed the question whether we can incorporate such an inference rule into our framework, and that was partly the motivation for developing this paper.

The critical difference is that global quantifiers $\tilde{\forall}x.F$ range over *arbitrary* random variables: functions from the sample space Ω to values. A random variable x can take different values on different traces, unlike local quantification where x denotes a specific value on the current trace.

A local formula ϕ is interpreted as a family of functions $\llbracket \phi \rrbracket_{\eta, \rho}$ from random tapes ρ to Boolean values.

As for global formulas, the overwhelming truth judgment $\llbracket \phi \rrbracket$ means:

$$\Pr_{\rho}[\llbracket \phi \rrbracket_{\eta, \rho} = \text{true}] = 1 - \text{negl}(\eta).$$

The rest of the global semantics is essentially the same as that of indistinguishability CCSA except that the domain of interpretation is arbitrary random variables, not PPT algorithms.

B. Higher-Order Features and Many-Sorted Logic

As we discussed, Squirrel’s logic is “higher-order” because quantification over function-typed variables is allowed, making them second-order over individual traces. However, *global* formulas—which is what matters for us as we only quantify globally—as long as soundness is concerned, Squirrel’s quantification is many-sorted first-order.

At the global level, Squirrel’s quantification is over random variables, which are functions from traces to values. When we write $\tilde{\forall}f : (\text{message} \rightarrow \text{message}).F$, the variable f ranges over random variables of function type—that is, functions that on each trace ρ yield a function from messages to messages. This is *many-sorted first-order logic* where:

- Types generate sorts: base types (message, bool, timestamp, index) plus function spaces $(\tau_1 \rightarrow \tau_2)$
- A function symbol $(_ _)$ is introduced for β -denotation: $(\lambda x.t \ u) = t(u/x)$.
- Each sort has a domain of interpretation: $\llbracket \tau_1 \rightarrow \tau_2 \rrbracket_{\eta} = \llbracket \tau_1 \rrbracket_{\eta} \rightarrow \llbracket \tau_2 \rrbracket_{\eta}$
- Application is a family of function symbols: one for each function type
- Quantifiers bind individual variables ranging over elements of sorts

This distinction is crucial: As we show in Section VIII, Fitting’s embedding applies directly to many-sorted first-order logic, including sorts that happen to be function spaces. This means all the benefits of our approach—cut-elimination, semantic clarity, proper handling of quantifiers—extend to Squirrel’s full type system.

C. Case Analysis and the *const* Predicate

As it is explained in [2]: During their proofs “one may write e.g. an authentication property as a formula of the form

$$[\forall \tau. \text{happens}(\tau) \Rightarrow \text{condition@}\tau \Rightarrow \exists \tau'. \tau' < \tau \wedge \text{event@}\tau']$$

which intuitively expresses that, in any execution trace of a protocol, with overwhelming probability, if some participant checks a condition at any time point inside the trace, some event must have happened earlier in the trace. [...] Such authentication formulas might be consequences of axioms

expressing (overwhelmingly true) cryptographic assumptions, e.g. about signatures, and they might have more complex formulas as consequences – the CCSA logics feature a predicate expressing computational indistinguishability”. In other words, the implication in the middle is not proven by trace-wise argument, but one has to carry out a global argument. If the protocol only has actions A and B , then this reduces to proving something like

$$\tilde{\forall}(\tau : \text{timestamp}). [\tau = A \vee \tau = B] \Rightarrow [\phi]$$

The problem is however that when Squirrel quantification is global, a witness can vary uncontrollably, depend on the future as it is any random variable. Antoine and Baelde note that this problem can be worked around by assuming τ is *constant* across traces, thereby preventing it to vary.

$$\tilde{\forall}(\tau : \text{timestamp}). \text{const}(\tau) \Rightarrow [\tau = A \vee \tau = B] \Rightarrow \phi$$

D. Squirrel’s Machinery for the *const* Predicate

The $\text{const}(t)$ predicate in Squirrel is needed to prevent the argument from varying from trace to trace badly. Since quantification goes trace-by-trace, they are forced to allow arbitrary random variables globally, but to keep control over bad behavior they introduce the $\text{const}(t)$ predicate that ensures good behavior over traces and inference rules for it. It asserts that term t evaluates to the same value for all security parameters η and random tapes ρ . This predicate serves two distinct purposes:

- 1) *Constant bitstrings*: A term like a timestamp variable x is constant if it denotes the same timestamp across all traces—effectively a ground value.
- 2) *Constant handles*: More subtly, $\text{const}(t)$ can mean that t corresponds to a single key generation or other honest sampling, even though the actual bitstring varies. For instance, $\text{const}(k)$ for a key name k means the term denotes the *same* sampling across different uses, ensuring it corresponds to one key generation rather than independent samplings on different traces.

Squirrel provides inference rules to establish constancy. The key rules include:

- G.CONST:APP: If t and t' are constant, then $t \ t'$ is constant.
- G.CONST:QUANT: For binders $Q \in \{\lambda, \forall, \exists\}$ over a fixed type τ : if for all constant x , the body t is constant, then $Q(x : \tau).t$ is constant.
- L.R- $\forall$ -CONST: An alternative universal quantifier introduction rule that adds $\text{const}(x)$ as a hypothesis when the type is finite and fixed.

Crucially, when moving in and out of $\llbracket _ \rrbracket$ with universal quantification, one would like to keep control of the behavior, so Proposition 4 of [5] establishes that for finite fixed types:

$$\mathcal{M} \models [\forall(x : \tau). \phi] \quad \text{iff} \quad \mathcal{M} \models \tilde{\forall}(x : \tau). \text{const}(x) \Rightarrow [\phi]$$

In the next section we show how our technique resolves these issues.

VIII. HOW FITTING’S EMBEDDING HELPS SQUIRREL

Without suggesting to introduce explicit modalities in Squirrel, we show how to simplify its syntax by keeping the $[\phi]$ notation but redefine its semantics as the semantics of ϕ^* —modalities appear only at the semantic level, as in BC twisted semantics. All Squirrel inference rules remain sound but can be simplified by removing the `const` predicate machinery.

a) *Many-sorted logic and “higher-order” features.*: As discussed in the previous section, for the sake of soundness—which is what we need for protocol verification—one can just treat quantification over functions as many-sorted first-order quantification. For example, consider a Squirrel formula with higher-order quantification: $[\forall f : (\text{message} \rightarrow \text{message}). \forall x : \text{message}. f(f(x)) = f(x) \rightarrow f(x) = x]$. This is many-sorted FOL with sorts `message` and $(\text{message} \rightarrow \text{message})$. The Fitting twist gives:

$$\forall f. \forall x. \Box((f(f(x)) = f(x))^* \rightarrow (f(x) = x)^*)$$

where $(t_1 = t_2)^* = \Box \Diamond(t_1 = t_2)$ for equations.

Therefore, all the benefits of the Fitting approach—cut-elimination, semantic clarity, proper handling of quantifiers—apply directly to Squirrel’s full term language. There is no need to develop separate machinery for “higher-order” features; they are accommodated by many-sorted first-order theory.

b) **No need for separate local and global reasoning.**: In Squirrel one has to go back and forth between local and global reasoning on a local formula ϕ . Local reasoning does not allow the use of cryptography as cryptography is inherently global. For global reasoning one goes to $[\phi]$ but then it is difficult to go back as we described earlier for case distinction in IV-0d: From $[\phi_1] \vdash [\psi]$ and $[\phi_2] \vdash [\psi]$ we are able to obtain $[\phi_1 \vee \phi_2] \vdash [\psi]$. In our semantics there is no going back and forth, everything can be done globally. If we do not want to introduce modalities in the syntax, just use the Fitting first-order soundness for the formulas of the form $[\phi]$ with global quantification.

c) **Universal quantification.**: In Squirrel, the relationship between local and global quantifiers requires the equivalence that only works in certain situations (Proposition 4 of [5]), namely for finite fixed finite types:

$$\mathcal{M} \models [\forall x. \phi] \iff \mathcal{M} \models \tilde{\forall} x. \text{const}(x) \Rightarrow [\phi]$$

With Fitting’s embedding, this always holds: $\forall x \phi^* \leftrightarrow (\forall x \phi)^*$. That is, as in our case $[\phi] \equiv \phi^*$:

$$\forall x[\phi] \leftrightarrow [\forall x \phi]^*$$

x can have whatever type we want. Something that is truly constant across traces, something that is PPT, or an arbitrary random variable.

d) **Case analysis without the `const` restriction.**: Returning to Section VII-C’s case distinction, the twisted semantics offers two solutions. Define the semantics of timestamps as constants. Then problematic “random variables over timestamps” (those that are A on some tapes and B on others)

simply do not exist in our quantification domain—quantifiers range over timestamps, not functions from sample space to timestamps.

e) **Honest nonce generation without the `const` restriction.**: Since global quantification over nonces, keys, etc can result in objects that vary uncontrollably from trace to trace, cryptographic axioms such as IND-CCA and others also need the `const` predicate to make sure that there is a single handle representing the given nonce, key, etc. This ensures that the nonce, key etc is really an honestly generated one. This too, becomes unnecessary with the BC twisted approach.

f) **Eliminating the `const` restriction.**: More broadly, the entire `const` infrastructure in Squirrel—the predicate itself, the type labels $\{\text{fixed}, \text{finite}\}$, and the restricted inference rules—likely becomes unnecessary in Fitting’s approach. Consider the key inference rules from [5] that require constancy conditions:

G.R-LOC: $\rightarrow$ (proving local implication):

$$\frac{E; \Theta, [\phi] \vdash [\psi] \quad E; \Theta \vdash \text{const}(\phi)}{E; \Theta \vdash [\phi \Rightarrow \psi]}$$

L.R- $\forall$ -CONST (universal introduction with constancy):

$$\frac{\{\text{fixed}, \text{finite}\} \subseteq \text{label}(\tau) \quad E, x : \tau; \Theta, \text{const}(x); \Gamma \vdash \phi}{E; \Theta; \Gamma \vdash \forall(x : \tau). \phi}$$

G.L-LOC: $\vee$ (case analysis on local disjunction):

$$\frac{E; \Theta, [\phi] \vdash F \quad E; \Theta, [\psi] \vdash F \quad E; \Theta \vdash \text{const}(\phi) \tilde{\vee} \text{const}(\psi)}{E; \Theta, [\phi \vee \psi] \vdash F}$$

In Fitting’s framework, the first two rules become standard first-order rules without side conditions:

Implication: Squirrel’s G.R-LOC: $\rightarrow$ requires `const`(ϕ) to prove $[\phi \Rightarrow \psi]$. However, if we define $[\phi]$ to be the Fitting twist ϕ^* , then the above rule is sound without the additional `const`(ϕ) assumption by Fitting’s embedding theorem.

Universal introduction: Squirrel’s L.R- $\forall$ -CONST requires the type to be $\{\text{fixed}, \text{finite}\}$ to gain the `const`(x) hypothesis. In Fitting, universal quantification is just standard $\forall$ over domain elements—no need to have a stronger universal quantification introduction with `const`(x) as a hypothesis, because variables always denote domain elements.

Case analysis: As long as F is a formula of the form $[\phi]$, this too is solved by Fitting. However, when F is any global formula, then additional condition is needed, namely that ϕ is determined.

g) **Determined formulas.**: Because of the aforementioned case distinction issues, Squirrel need to deal with determined formulas: those for which $[\phi] \vee [\neg \phi]$ holds—it is either overwhelmingly true or overwhelmingly false. With some additional inference rules on these and the constant predicate they are able to overcome the issues that they cannot patch together the case distinctions easily. Should determined formulas still be needed with our semantics, they are readily available, also in the form of $[\phi] \vee [\neg \phi]$, but now this means $\phi^* \vee (\neg \phi)^*$.

h) **The twisted semantics existential’s varying witnesses.**: An important point concerns Fitting’s treatment of ex-

istential quantification. The embedding $(\exists x.\phi)^* = \Box\Diamond\exists x.\phi^*$ does *not* require a single witness that works across all traces with overwhelming probability. Instead, it means: for all non-negligible sets S' , there exists a non-negligible subset $S'' \subseteq S'$ such that $\exists x.\phi^*$ holds throughout S'' . Such a witness is a global object satisfying something locally on a non-negligible set.

IX. TWISTED SEMANTICS FOR CRYPTOVAMPIRE'S EVALUATED LOGIC

The purpose of this section is the following: On the one hand, CryptoVampire uses bounded Skolemization to avoid nonce-guessing oracle functions. We show how this issue does not emerge with the original BC twisted semantics, basically because you would have to guess nonces on non-negligible sets (not on traces), which you cannot do. On the other hand we also show how CryptoVampire's hybrid model—computational model combined with symbolic—works with the twisted semantics. We would like to emphasize that CryptoVampire basic syntax would not need to change, it stays first-order, no need for S4. Getting rid of Bounded Skolemization would be entirely for free.

CryptoVampire [25] automates proofs in the BC Logic framework by encoding it into first-order logic for saturation-based theorem provers. Notably, CryptoVampire focuses on *trace properties* rather than equivalence properties, which means the indistinguishability predicate $\sim$ is not involved. This makes CryptoVampire's setting a natural match for Bana and Comon's original twisted semantics, since the original BC reachability framework (Section II-B) was designed precisely for trace properties (with computability).

CryptoVampire's *Evaluated Logic* elegantly separates symbolic reasoning from computational reasoning through an evaluation predicate. However, the computational semantics follows a trace-by-trace approach, requiring complex bounded Skolem machinery to prevent unsoundness. We propose retaining CryptoVampire's symbolic structure while replacing the trace-by-trace semantics with the BC twisted semantics.

Note that no modalities appear in the syntax: we do not suggest that CryptoVampire should use modalities or change in any way.

A. CryptoVampire's Evaluated Logic

CryptoVampire works with a term algebra where terms represent symbolic computations. The key innovation is distinguishing between *symbolic* predicates (which depend only on term structure) and *evaluated* predicates (which depend on computational execution). Subterm relation for them is an axiomatized predicate, but as subterms cannot be checked on bit strings, they pair the computational model with a symbolic one built on ground terms.

a) *Syntax*.: The Evaluated Logic has two layers:

Symbolic terms are built from function symbols, nonces, and variables:

$$t ::= x \mid n[\vec{z}] \mid f[\vec{z}](\vec{t}) \mid \text{input}(T) \mid \dots$$

where $n[\vec{z}]$ are indexed nonces (representing fresh randomness), $f[\vec{z}]$ are indexed function symbols, and $\text{input}(T)$ represents attacker input at step T .

Formulas combine symbolic predicates with an evaluation predicate:

- Symbolic equality: $t = t'$ (syntactic identity of terms)
- Subterm relation: $t \sqsubseteq t'$ (whether t appears syntactically in t')
- Evaluation predicate: $|t|$ for Boolean-sorted terms t
- Standard first-order connectives and quantifiers

b) *The EQ function symbol*.: A crucial point for understanding CryptoVampire's notation is the treatment of computational equality. The notation $|t| = |t'|$ is *syntactic sugar* for $|\text{EQ}(t, t')|$, where EQ is a Boolean-sorted function symbol in the term algebra as before. That is:

$$|t| = |t'| \text{ abbreviates } |\text{EQ}(t, t')|$$

The function symbol EQ (written $\equiv$ in CryptoVampire's notation) takes two message-sorted terms and produces a Boolean-sorted term. On execution, $\text{EQ}(t, t')$ evaluates to 1 if t and t' compute to the same bitstring, and 0 otherwise.

c) *The Symbolic Model*.: CryptoVampire introduces a *symbolic model* M that interprets symbolic predicates over ground terms independently of random tapes:

- $M(=)(t, t')$ holds iff t and t' are syntactically identical terms
- $M(\sqsubseteq)(t, t')$ holds iff t appears as a subterm of t'

These are purely structural properties of the term algebra, determined without any computational execution.

d) *CryptoVampire's Trace-by-Trace Semantics*.: The original semantics (Definition 6 and Figure 3 of [25]) defines a *cryptographic model* $\mathcal{L} = (\mathcal{M}, \mathbb{T}, M)$ consisting of a computational model $\mathcal{M}$, a trace $\mathbb{T}$, and a symbolic model M . Following the BC Logic, each computation uses a security parameter η and a pair of random tapes $\rho = (\rho_h, \rho_A)$ —one for honest agents and one for the adversary. Nonces are slices of the honest tape ρ_h . The interpretation is:

$$\begin{aligned} \mathcal{L}^\eta(t \sqsubseteq t')(\rho) &:= M(\sqsubseteq)(t, t') && \text{(symbolic, independent of } \rho) \\ \mathcal{L}^\eta(t = t')(\rho) &:= M(=)(t, t') && \text{(symbolic, independent of } \rho) \\ \mathcal{L}^\eta(|t|)(\rho) &:= \min(1, \llbracket [t]^\mathbb{T} \rrbracket_{\eta, \rho}) && \text{(evaluated trace-by-trace)} \end{aligned}$$

where $[t]^\mathbb{T}$ unfolds the symbolic term t according to trace $\mathbb{T}$ into a BC Logic term, and $\llbracket \cdot \rrbracket_{\eta, \rho}$ denotes the computational interpretation given by $\mathcal{M}$.

Cryptographic satisfiability (Definition 7 of [25]) requires:

$$\mathcal{L} \models_P \phi \text{ when } \Pr_{\rho}(\mathcal{L}^\eta(\phi)(\rho) \neq 1) \leq \text{negl}(\eta)$$

As Remark 5 of [25] notes, the evaluation predicate $|t|$ is “defined *trace-wise*”—it evaluates to 0 or 1 on each individual tape. This creates the fundamental issue we now address.

B. The Nonce Guessing Problem

The trace-by-trace semantics creates issues with existential quantification over terms. CryptoVampire's Theorem 2 (No

Guessing) states that for any ground term m :

$$|n[\bar{z}]| = |m| \Rightarrow n[\bar{z}] \sqsubseteq m$$

This axiom is cryptographically valid: if the ground term m evaluates to the same bitstring as the nonce $n[\bar{z}]$, then m must contain $n[\bar{z}]$ syntactically since guessing a random nonce is negligible from terms that are not created using the nonce. Note that in their semantics the domain is not arbitrary PPT algorithms, but only those that are created from closed terms.

However, Example 16 of [25] shows the universally quantified version fails:

$$\forall x. (|n[\bar{z}]| = |x| \Rightarrow n[\bar{z}] \sqsubseteq x)$$

is invalid in trace-by-trace semantics. The problem is as follows. In trace-by-trace semantics, evaluating $\forall x. \phi$ on random tape ρ requires ϕ to hold for all ground terms x . On one fixed string ρ_1 on the tape, some term t_1 might happen to match the nonce value; on another fixed string ρ_2 on the tape, a different term t_2 matches. When the formula is Skolemized to prove the negation, we get a Skolem function sk such that $|n[\bar{z}]| = |\text{sk}|$ holds on every random tape—but sk is a non-computable oracle that “guesses” the nonce value differently on each tape.

Example 17 of [25] makes this concrete: assuming the term algebra contains 0 and successor $S(\cdot)$, for any random tape $\rho = (\rho_h, \rho_A)$, the nonce $n[\bar{z}]$ (which is a slice of the honest tape ρ_h) evaluates to some natural number N , and the term $S^N(0)$ evaluates to the same value. The Skolem witness can pick this term differently for each ρ , violating the no-guessing principle.

CryptoVampire addresses this through *bounded Skolem normal forms* (Definition 18), restricting Skolemization to ensure witnesses come from a finite syntactic domain. This is effective but adds considerable complexity to the metatheory.

C. The BC Twisted Semantics for CryptoVampire

Since CryptoVampire handles only trace properties (not equivalence), the original BC reachability semantics (Section II-B) applies directly. We propose keeping CryptoVampire’s symbolic structure while replacing the trace-by-trace semantics with the BC twisted semantics. To emphasize that CryptoVampire’s language remains purely first-order, we define the semantics *directly*, without modal operators.

Definition 9.1 (BC Twisted Semantics for CryptoVampire): Let $\mathcal{M}^C$ be a computational model, M a symbolic model, S a non-negligible subset of traces, and $\nu : \text{Var} \rightarrow D$ a valuation into the domain D of ground terms.

Symbolic predicates (trace-independent, interpreted by M):

- $(\mathcal{M}^C, M), S, \nu \models_{BC} (t = t')$ iff $M(=)(\nu(t), \nu(t'))$
- $(\mathcal{M}^C, M), S, \nu \models_{BC} (t \sqsubseteq t')$ iff $M(\sqsubseteq)(\nu(t), \nu(t'))$

These hold independently of S —if they hold at S , they hold at any $S' \subseteq S$ so we do not need the box diamond pattern.

Evaluation predicate (trace property, requires overwhelming satisfaction):

- $(\mathcal{M}^C, M), S, \nu \models_{BC} |t|$ iff for all non-negligible $S' \subseteq S$, there exists a non-negligible $S'' \subseteq S'$ such that for all η and all $\rho \in S''_\eta$, $\llbracket \nu(t) \rrbracket_{\eta, \rho}^T = 1$. This is the same as requiring $\Pr_\rho(\rho \in S : \llbracket \nu(t) \rrbracket_{\eta, \rho}^T = 1)$ is overwhelming.

In other words, $|t|$ is satisfied iff t evaluates to 1 with overwhelming probability on S .

Propositional connectives, quantification and validity works exactly as in Definition 2.2.

Observe that the symbolic predicates ($=, \sqsubseteq$) are *persistent*—they depend only on term structure, not on traces—so the “for all S' , exists S'' ” pattern is equivalent with not having them there. Only the evaluation predicate $|t|$ and the logical connectives that can “hide” it (disjunction, negation, implication, existential) require the characteristic BC twisted pattern. Here too we have If $\Gamma \vdash_{\text{FOL}} \phi$ in classical first-order logic, then:

$$(\mathcal{M}^C, M) \models_{BC} \Gamma \implies (\mathcal{M}^C, M) \models_{BC} \phi$$

which follows from the general soundness of the BC twisted semantics (Section II-B). The key point is that CryptoVampire can use standard first-order theorem provers without modification—the non-standard semantics ensures soundness automatically.

D. Resolution of the Nonce Guessing Problem

Consider the No Guessing axiom $\forall x. (|n[\bar{z}]| = |x| \Rightarrow n[\bar{z}] \sqsubseteq x)$ under the BC twisted semantics (Definition 9.1).

Proposition 9.2: The formula $\forall x. (|n[\bar{z}]| = |x| \Rightarrow n[\bar{z}] \sqsubseteq x)$ is valid in the BC twisted semantics.

Proof: By Definition 9.1, validity means: for all ground terms $t \in D$ and all non-negligible sets S ,

$$(\mathcal{M}^C, M), S \models_{BC} |n[\bar{z}]| = |t| \Rightarrow n[\bar{z}] \sqsubseteq t$$

By the clause for implication, this means: for all non-negligible $S' \subseteq S$, if $(\mathcal{M}^C, M), S' \models_{BC} |n[\bar{z}]| = |t|$, then $(\mathcal{M}^C, M), S' \models_{BC} n[\bar{z}] \sqsubseteq t$.

Now suppose $n[\bar{z}] \not\sqsubseteq t$, i.e., the nonce does not appear syntactically in t . Then t ’s value is computed independently of the nonce’s random bits. By the clause for the evaluation predicate, $|n[\bar{z}]| = |t|$ (i.e., $|\text{EQ}(n[\bar{z}], t)|$) being satisfied at S' would mean that $|\text{EQ}(n[\bar{z}], t)|$ is satisfied overwhelmingly on S' . But this is impossible because since t is independent of $n[\bar{z}]$, the probability that they match on any set is negligible.

Therefore, if $n[\bar{z}] \not\sqsubseteq t$, then $(\mathcal{M}^C, M), S' \not\models_{BC} |n[\bar{z}]| = |t|$ for any non-negligible S' , making the implication true. $\square$

X. CONCLUSION

We have presented a unified semantic framework for reasoning about cryptographic protocol security that bridges the gap between trace properties and global properties like computational indistinguishability. Our approach resolves two open problems identified by Antoine and Baelde [2]: extending overwhelming truth to first-order quantification, and unifying it with indistinguishability-based reasoning.

The key insight is that the original “twisted” computational semantics of Bana and Comon [10], when understood through Fitting’s embedding into S4 modal logic, provides exactly the right framework.

We established that first-order S4 with the persistence axiom (FOS4_P) is sound and complete for our computational semantics, and showed that for quantifier-free formulas, our semantics coincides with Antoine and Baelde’s S5-based overwhelming truth. For first-order formulas over random variable domains, our semantics matches Squirrel’s trace-by-trace evaluation.

Our framework offers practical simplifications for existing tools for free. For the Squirrel prover, the Fitting embedding eliminates the need for the `const(x)` predicate and its associated restrictions—constancy across traces becomes an inherent property of rigid designators in constant-domain S4. For CryptoVampire, we proposed a hybrid semantics that retains symbolic term algebra while replacing trace-by-trace evaluation with the BC twisted semantics, eliminating the need for bounded Skolemization entirely.

The move from S5 to S4 also provides proof-theoretic advantages: S4 admits cut-elimination in standard sequent calculus, enabling cleaner metatheoretic analysis.

Currently, our technique works well combining overwhelming satisfaction of trace properties and indistinguishability at the global level, but to argue effectively with explicit S4 modalities about cryptographic properties remains future work.

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APPENDIX

A. Modal Logic Background

We provide a brief overview of modal logic concepts used in this paper. For comprehensive treatments, see [16], [24].

a) *Propositional modal logic.*: Modal logic extends classical propositional logic with operators $\Box$ (necessity) and $\Diamond$ (possibility), where $\Diamond\phi \equiv \neg\Box\neg\phi$. The semantics uses *Kripke frames*: a set W of *worlds* with an *accessibility relation* $R \subseteq W \times W$. A *Kripke model* adds a valuation V assigning propositions to worlds. Truth is defined:

- $\mathcal{M}, w \models \Box\phi$ iff for all w' with wRw' , $\mathcal{M}, w' \models \phi$

- $\mathcal{M}, w \models \Diamond\phi$ iff for some w' with wRw' , $\mathcal{M}, w' \models \phi$
- b) *S4 and S5*: Different axiom systems correspond to different constraints on R :
 - **K**: The minimal normal modal logic. Axiom: $\Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$.
 - **T**: K + reflexivity (wRw). Axiom: $\Box\phi \rightarrow \phi$.
 - **S4**: T + transitivity ($wRw' \wedge w'Rw'' \Rightarrow wRw''$). Axiom: $\Box\phi \rightarrow \Box\Box\phi$.
 - **S5**: S4 + symmetry ($wRw' \Rightarrow w'Rw$), equivalently T + Euclidean ($wRw' \wedge wRw'' \Rightarrow w'Rw''$). Axiom: $\Diamond\phi \rightarrow \Box\Diamond\phi$.

S5's equivalence relation makes all worlds mutually accessible from any world, collapsing modal distinctions: $\Box\Diamond\phi \leftrightarrow \Diamond\phi$ and $\Diamond\Box\phi \leftrightarrow \Box\phi$ in S5, but not in S4.

c) *First-order modal logic*: Extending to first-order requires decisions about the domain. In *constant domain* semantics, all worlds share the same domain D . This validates the *Barcan formula* $\forall x.\Box\phi \rightarrow \Box\forall x.\phi$ and its converse. With constant domains, quantifiers commute with $\Box$: one quantifies over domain elements that exist “necessarily.”

d) *Fitting's embedding*: Fitting [24] showed that intuitionistic logic embeds into S4 via the translation $\phi \mapsto \phi^*$ where atomic $P^* = \Box\Diamond P$, $(\phi \wedge \psi)^* = \phi^* \wedge \psi^*$, $(\neg\phi)^* = \Box\neg\phi^*$, $(\phi \vee \psi)^* = \Box(\Diamond\phi^* \vee \Diamond\psi^*)$, $(\forall x.\phi)^* = \forall x.\phi^*$, and $(\exists x.\phi)^* = \Box\Diamond\exists x.\phi^*$. This embedding is *faithful*: ϕ is intuitionistically valid iff ϕ^* is S4-valid. The $\Box\Diamond$ pattern for atoms ensures “stability”—truth persists to accessible worlds.

e) *Relevance to this paper*: Our computational Kripke models use non-negligible sets of traces as worlds, with $S' \subseteq S$ as accessibility. This yields S4 (reflexive, transitive) but not S5 (not symmetric: $S' \subseteq S$ does not imply $S \subseteq S'$). Fitting's embedding translates overwhelming truth ($\Box\Diamond P$ = “for all non-negligible subsets, P holds on some non-negligible subset”) into S4, providing cut-elimination and clean quantifier semantics unavailable in S5-based approaches.

B. Equivalence Proofs

We provide the full proofs of the equivalence results from Section II.

Proposition 2.4.[*Equivalence for Quantifier-Free Formulas*] *Let ϕ be a quantifier-free formula. Then ϕ holds computationally (i.e., $\mathcal{M}^C \models_{BC} \phi$) if and only if ϕ holds with overwhelming probability in the sense of [2].*

Proof: By definition, $\mathcal{M}^C \models_{BC} \phi$ iff $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \phi^*$ where ϕ^* is the Fitting twist. We proceed by structural induction on ϕ .

Base case (atomic formulas): For an atomic formula P , the Fitting twist gives $P^* = \Box\Diamond P$. We show:

$$\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \Box\Diamond P \iff$$

P holds with overwhelming probability

($\Rightarrow$) Suppose $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \Box\Diamond P$ but P fails on a non-negligible set S_0 . By the $\Box$ operator, $\Diamond P$ holds at S_0 , so there exists non-negligible $S'_0 \subseteq S_0$ where P holds throughout—contradicting that P fails throughout S_0 .

($\Leftarrow$) Suppose P holds except on negligible set N . For any non-negligible S' , take $S'' = S' \setminus N$. Since N is negligible and S' is non-negligible, S'' is non-negligible and P holds throughout S'' . Thus $\Diamond P$ holds at every non-negligible S' , giving $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \Box\Diamond P$.

Conjunction: $(\phi_1 \wedge \phi_2)^* = \phi_1^* \wedge \phi_2^*$. By induction, $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \phi_i^*$ iff ϕ_i holds overwhelmingly. Since the intersection of two sets of overwhelming measure has overwhelming measure, $\phi_1 \wedge \phi_2$ holds overwhelmingly iff both ϕ_1 and ϕ_2 do.

Negation: $(\neg\psi)^* = \Box\neg\psi^*$. The formula $\neg\psi$ holds overwhelmingly means ψ holds only on a negligible set. For any non-negligible S' , the set where ψ holds within S' is negligible, so ψ^* fails at S' (by induction), giving $\neg\psi^*$ at S' . Thus $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \Box\neg\psi^*$ holds. The converse is similar.

Implication and disjunction: These follow from the above cases using the standard equivalences $(\phi \rightarrow \psi) \leftrightarrow (\neg\phi \vee \psi)$ and De Morgan's laws which are sound because of Fitting's theorem. $\square$

Theorem 2.7.[*Equivalence for Random Variable Domains*] *Let $\mathcal{M}^C$ be a computational model with random variable domain D , satisfying Assumption 1. For any first-order formula ϕ and valuation ν :*

$$\mathcal{M}^C, \Omega^C, \nu \models_{BC} \phi \iff$$

ϕ holds trace-by-trace with overwhelming probability

Proof: By definition, $\mathcal{M}^C, \Omega^C, \nu \models_{BC} \phi$ iff $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \phi^*$. We proceed by structural induction on ϕ . Throughout, Assumption 1 ensures that all sets of traces we consider are measurable.

Base case (atomic formulas): Identical to Proposition 2.4.

Connectives ($\neg, \wedge, \vee, \rightarrow$): Identical to Proposition 2.4.

Existential quantification: Let $\phi = \exists x.\psi$. Then $(\exists x.\psi)^* = \Box\Diamond\exists x.\psi^*$.

($\Leftarrow$) Suppose $\exists x.\psi$ holds trace-by-trace with overwhelming probability. Let $S_{\text{good}} = \{\rho \mid \exists \mathbf{a} \in D. \psi \text{ holds at } \rho \text{ under } \nu[x \mapsto \mathbf{a}]\}$. Then $\mu(S_{\text{good}})$ is overwhelming.

For each $\rho \in S_{\text{good}}$, there is some witnessing value $v_\rho \in V$ such that ψ holds at ρ when x is assigned v_ρ . Define a new random variable $\mathbf{a}^* : \Omega^C \rightarrow V$ by setting $\mathbf{a}^*(\rho) = v_\rho$ for $\rho \in S_{\text{good}}$ (and arbitrary values elsewhere). Since D consists of all measurable functions $\Omega^C \rightarrow V$, we have $\mathbf{a}^* \in D$.

By construction, ψ holds at every $\rho \in S_{\text{good}}$ under $\nu[x \mapsto \mathbf{a}^*]$. Since S_{good} has overwhelming measure, ψ holds trace-by-trace with overwhelming probability under $\nu[x \mapsto \mathbf{a}^*]$. By induction, $\mathcal{M}^{KC}, \Omega^C, \nu[x \mapsto \mathbf{a}^*] \models_{S4} \psi^*$, and therefore $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \exists x.\psi^*$.

Now we must show $\Box\Diamond\exists x.\psi^*$ holds. Let S be any non-negligible subset of Ω^C . We need to find a non-negligible $S' \subseteq S$ where $\exists x.\psi^*$ holds. Take $S' = S \cap S_{\text{good}}$. Since S is non-negligible and S_{good} has overwhelming measure (so its complement is negligible), S' is non-negligible. The witness $\mathbf{a}^*$ restricted to S' shows that $\exists x.\psi^*$ holds at S' .

($\Rightarrow$) Suppose $\mathcal{M}^C, \Omega^C, \nu \models_{BC} \phi$, i.e., $\mathcal{M}^{KC}, \Omega^C, \nu \models_{S4} \Box \Diamond \exists x. \psi^*$. We prove by contradiction that $\exists x. \psi$ holds trace-by-trace with overwhelming probability.

Suppose not. Then the set $N = \{\rho \mid \forall \alpha \in D. \psi \text{ fails at } \rho \text{ under } \nu[x \mapsto \alpha]\}$ is non-negligible. Since $\Box \Diamond \exists x. \psi^*$ holds at Ω^C , in particular $\Diamond \exists x. \psi^*$ holds at N . This means there exists a non-negligible $S \subseteq N$ where $\exists x. \psi^*$ holds, i.e., there exists $\alpha \in D$ such that $\mathcal{M}^{KC}, S, \nu[x \mapsto \alpha] \models_{S4} \psi^*$.

By induction, ψ holds trace-by-trace with overwhelming probability (relative to S) under $\nu[x \mapsto \alpha]$. But every $\rho \in S \subseteq N$ is a trace where ψ fails under $\nu[x \mapsto \alpha]$ (by definition of N). This contradicts that ψ holds for overwhelming $\rho \in S$.

Universal quantification: Follows from the existential case using the equivalence $\forall x. \psi \equiv \neg \exists x. \neg \psi$ and the fact that Fitting's embedding preserves this equivalence. $\square$

We restate and prove the main results from Section III-A.

Theorem 3.2. [Propositional Completeness] *A formula ϕ is provable in $S4_P$ if and only if it is valid in all finite rooted tree models.*

Proof: Soundness: We verify that all axioms of $S4_P$ are valid in finite rooted tree models and that the inference rules preserve validity.

The $S4$ axioms are:

- (K) $\Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$: Valid in all Kripke frames.
- (T) $\Box\phi \rightarrow \phi$: Valid because R is reflexive.
- (4) $\Box\phi \rightarrow \Box\Box\phi$: Valid because R is transitive.

The persistence axiom $P \rightarrow \Box P$ is valid: if $w \in V(P)$ and wRu , then $u \in V(P)$ by the persistence condition on V . Hence if P is true at w , it is true at all accessible worlds, so $\Box P$ is true at w .

Modus ponens and necessitation preserve validity in all Kripke frames.

Completeness: Suppose $\not\models_{S4_P} \phi$. We construct a finite rooted tree model where ϕ fails.

Step 1: Canonical Model. Let $\mathcal{M}^c = (W^c, R^c, V^c)$ be the canonical model for $S4_P$:

- W^c is the set of all maximally $S4_P$ -consistent sets of formulas
- $\Gamma R^c \Delta$ iff for all $\Box\psi \in \Gamma$, we have $\psi \in \Delta$
- $\Gamma \in V^c(P)$ iff $P \in \Gamma$

The canonical model satisfies: (i) R^c is reflexive and transitive (standard for $S4$), and (ii) V^c is persistent. For (ii): if $\Gamma \in V^c(P)$ and $\Gamma R^c \Delta$, then $P \in \Gamma$. By the persistence axiom, $\Box P \in \Gamma$ (since Γ is maximally consistent and $P \rightarrow \Box P$ is an axiom). By definition of R^c , $P \in \Delta$, so $\Delta \in V^c(P)$.

Since $\not\models \phi$, the set $\{\neg\phi\}$ is consistent, so it extends to some $\Gamma_0 \in W^c$ with $\neg\phi \in \Gamma_0$.

Step 2: Filtration. Let Σ be the set of all subformulas of ϕ , closed under single negation (i.e., if $\psi \in \Sigma$ and ψ is not a negation, then $\neg\psi \in \Sigma$). This is a finite set. Define the equivalence relation on W^c :

$$\Gamma \sim \Delta \iff \forall \psi \in \Sigma : (\psi \in \Gamma \iff \psi \in \Delta)$$

The filtration $\mathcal{M}^f = (W^f, R^f, V^f)$ is defined by:

- $W^f = W^c / \sim$ (equivalence classes)
- $[\Gamma] R^f [\Delta]$ iff there exist $\Gamma' \in [\Gamma]$ and $\Delta' \in [\Delta]$ with $\Gamma' R^c \Delta'$
- $[\Gamma] \in V^f(P)$ iff $P \in \Gamma$ (for $P \in \Sigma$; well-defined since $\Gamma \sim \Gamma'$ implies $P \in \Gamma \iff P \in \Gamma'$)

Since $|\Sigma|$ is finite, there are at most $2^{|\Sigma|}$ equivalence classes, so W^f is finite.

Lemma A.1 (Filtration Preservation): For all $\psi \in \Sigma$ and all $\Gamma \in W^c$:

$$\psi \in \Gamma \iff \mathcal{M}^f, [\Gamma] \models \psi$$

Proof: By induction on the structure of ψ .

Base case ($\psi = P$ atomic): $P \in \Gamma$ iff $[\Gamma] \in V^f(P)$ iff $\mathcal{M}^f, [\Gamma] \models P$, by definition of V^f .

Negation ($\psi = \neg\chi$): $\neg\chi \in \Gamma$ iff $\chi \notin \Gamma$ (since Γ is maximally consistent) iff $\mathcal{M}^f, [\Gamma] \not\models \chi$ (by IH, since $\chi \in \Sigma$) iff $\mathcal{M}^f, [\Gamma] \models \neg\chi$.

Conjunction ($\psi = \chi_1 \wedge \chi_2$): $\chi_1 \wedge \chi_2 \in \Gamma$ iff $\chi_1 \in \Gamma$ and $\chi_2 \in \Gamma$ (maximal consistency) iff $\mathcal{M}^f, [\Gamma] \models \chi_1$ and $\mathcal{M}^f, [\Gamma] \models \chi_2$ (by IH) iff $\mathcal{M}^f, [\Gamma] \models \chi_1 \wedge \chi_2$.

Box ($\psi = \Box\chi$):

($\Rightarrow$) Suppose $\Box\chi \in \Gamma$. Let $[\Gamma] R^f [\Delta]$. Then there exist $\Gamma' \sim \Gamma$ and $\Delta' \sim \Delta$ with $\Gamma' R^c \Delta'$. Since $\Box\chi \in \Gamma$ and $\Box\chi \in \Sigma$, we have $\Box\chi \in \Gamma'$. By definition of R^c , $\chi \in \Delta'$. Since $\chi \in \Sigma$ and $\Delta' \sim \Delta$, we have $\chi \in \Delta$. By IH, $\mathcal{M}^f, [\Delta] \models \chi$. Since $[\Delta]$ was arbitrary with $[\Gamma] R^f [\Delta]$, we have $\mathcal{M}^f, [\Gamma] \models \Box\chi$.

($\Leftarrow$) Suppose $\Box\chi \notin \Gamma$. We must find $[\Delta]$ with $[\Gamma] R^f [\Delta]$ and $\mathcal{M}^f, [\Delta] \not\models \chi$.

Since $\Box\chi \notin \Gamma$ and Γ is maximally consistent in $S4_P$, the set $\{\theta : \Box\theta \in \Gamma\} \cup \{\neg\chi\}$ is consistent. Extend it to a maximally consistent Δ . Then $\Gamma R^c \Delta$ and $\neg\chi \in \Delta$, so $\chi \notin \Delta$. By IH, $\mathcal{M}^f, [\Delta] \not\models \chi$. Since $\Gamma R^c \Delta$, we have $[\Gamma] R^f [\Delta]$. $\square$

The filtration preserves reflexivity and transitivity (standard). For persistence: we need to ensure $\Box P \in \Sigma$ whenever $P \in \Sigma$ is atomic. We achieve this by closing Σ under the operation: if $P \in \Sigma$ is atomic, add $\Box P$ to Σ . This keeps Σ finite.

Now suppose $[\Gamma] \in V^f(P)$ and $[\Gamma] R^f [\Delta]$. Then $P \in \Gamma$, so $\Box P \in \Gamma$ (by the persistence axiom and maximal consistency). There exist $\Gamma' \sim \Gamma$ and $\Delta' \sim \Delta$ with $\Gamma' R^c \Delta'$. Since $\Box P \in \Sigma$ and $\Gamma \sim \Gamma'$, we have $\Box P \in \Gamma'$. By definition of R^c , $P \in \Delta'$. Since $P \in \Sigma$ and $\Delta' \sim \Delta$, we have $P \in \Delta$, giving $[\Delta] \in V^f(P)$.

Step 3: Unraveling into a Tree. We unravel $\mathcal{M}^f$ into a tree $\mathcal{T} = (W^{\mathcal{T}}, R^{\mathcal{T}}, V^{\mathcal{T}})$ rooted at $[\Gamma_0]$:

- $W^{\mathcal{T}}$ consists of finite paths $w_0 R^f w_1 R^f \dots R^f w_n$ starting from $w_0 = [\Gamma_0]$
- $(w_0, \dots, w_n) R^{\mathcal{T}} (w_0, \dots, w_n, w_{n+1})$ and reflexively $(w_0, \dots, w_n) R^{\mathcal{T}} (w_0, \dots, w_n)$
- $R^{\mathcal{T}}$ is closed under transitivity
- $(w_0, \dots, w_n) \in V^{\mathcal{T}}(P)$ iff $w_n \in V^f(P)$

The map $p : (w_0, \dots, w_n) \mapsto w_n$ is a bounded morphism, so truth is preserved: $\mathcal{T}, (w_0, \dots, w_n) \models \psi$ iff $\mathcal{M}^f, w_n \models \psi$ for $\psi \in \Sigma$.

The valuation $V^\mathcal{T}$ is persistent: if $(w_0, \dots, w_n) \in V^\mathcal{T}(P)$ and $(w_0, \dots, w_n)R^\mathcal{T}(w_0, \dots, w_m)$, then $w_n R^f w_m$ (by transitivity in $\mathcal{M}^f$), so $w_m \in V^f(P)$ by persistence in $\mathcal{M}^f$, giving $(w_0, \dots, w_m) \in V^\mathcal{T}(P)$.

Step 4: Finiteness via Bounded Modal Depth. The unraveling may be infinite, but we can truncate it using the following key observation:

Lemma A.2 (Modal Depth Locality): Let ψ be a formula of modal depth d , and let $\mathcal{M} = (W, R, V)$ be any Kripke model. For any world $w \in W$, the truth value of ψ at w depends only on the submodel induced by worlds reachable from w in at most d steps.

More precisely: if $\mathcal{M}' = (W', R', V')$ is another model and there is a relation $Z \subseteq W \times W'$ such that wZw' and Z is a d -bisimulation (bisimulation up to depth d), then $\mathcal{M}, w \models \psi$ iff $\mathcal{M}', w' \models \psi$.

Proof: By induction on ψ .

Base case (atomic P): Modal depth is 0. A 0-bisimulation requires $w \in V(P)$ iff $w' \in V'(P)$, which gives the result directly.

Boolean connectives: Immediate from the IH, as they don't increase modal depth.

Box ($\psi = \Box\chi$ where χ has depth $d-1$): The formula ψ has depth d . A d -bisimulation Z with wZw' satisfies:

- (Forth) If wRu , there exists u' with $w'R'u'$ and uZu' is a $(d-1)$ -bisimulation
- (Back) If $w'R'u'$, there exists u with wRu and uZu' is a $(d-1)$ -bisimulation

($\Rightarrow$) Suppose $\mathcal{M}, w \models \Box\chi$. Let $w'R'u'$. By (Back), there exists u with wRu and uZu' . Since $\mathcal{M}, w \models \Box\chi$, we have $\mathcal{M}, u \models \chi$. By IH (as χ has depth $d-1$), $\mathcal{M}', u' \models \chi$. Since u' was arbitrary, $\mathcal{M}', w' \models \Box\chi$.

($\Leftarrow$) Symmetric using (Forth). $\square$

Now define $\mathcal{T}_d$ by keeping only paths of length $\leq d$:

$$W^{\mathcal{T}_d} = \{(w_0, \dots, w_n) \in W^\mathcal{T} : n \leq d\}$$

with $R^{\mathcal{T}_d}$ and $V^{\mathcal{T}_d}$ inherited from $\mathcal{T}$.

The projection $p : (w_0, \dots, w_n) \mapsto w_n$ establishes a d -bisimulation between $\mathcal{T}_d$ at the root $([\Gamma_0])$ and $\mathcal{M}^f$ at $[\Gamma_0]$: the (Forth) and (Back) conditions hold because the unraveling faithfully represents all R^f -successors up to depth d .

By the Modal Depth Locality Lemma, $\mathcal{T}_d, ([\Gamma_0]) \models \phi$ iff $\mathcal{M}^f, [\Gamma_0] \models \phi$.

Since W^f is finite and paths have length $\leq d$, $\mathcal{T}_d$ is finite (at most $|W^f|^{d+1}$ nodes).

Conclusion. $\mathcal{T}_d$ is a finite rooted tree model (reflexive, transitive, persistent valuation) where ϕ fails at the root, since $\mathcal{T}_d, ([\Gamma_0]) \models \neg\phi$. $\square$

Proposition 3.3.[Computational Realizability] Any finite rooted tree model $\mathcal{M} = (W, R, V)$ can be realized by a computational model $\mathcal{M}^c = (\Omega^c, D, \cdot^{\mathcal{M}^c})$ where worlds correspond to non-negligible sets of traces.

Proof: Let L be the set of leaves of $\mathcal{M}$. For each security parameter η , we partition the sample space Ω_η^c into $|L|$ disjoint

sets $\{S_{l,\eta}\}_{l \in L}$, each with measure $1/|L|$. Since $|L|$ is fixed (independent of η), each $S_l = \{S_{l,\eta}\}_\eta$ is a non-negligible ensemble.

For the root w_0 , we set $S_{w_0} = \Omega^c$ (the full sample space). For any internal node w , define $S_w = \bigcup \{S_l \mid l \text{ is a descendant of } w\}$.

The valuation is defined as follows: for a trace $\rho \in S_l$ and atom P , we set P to hold at ρ iff $l \in V(P)$ in the original model $\mathcal{M}$. By the persistence property of $\mathcal{M}$, if P is true at w , it is true at all descendant leaves, and thus P holds for all traces in S_w .

The accessibility relation is realized by reverse inclusion: $S_w R S_{w'}$ iff $S_{w'} \subseteq S_w$, which holds iff w' is a descendant of w in the tree. $\square$

Theorem 3.5.[First-Order Completeness] A formula ϕ is provable in FOS4_P if and only if it is valid in all tree models with constant domains.

Proof: Soundness: The Barcan formulas are valid in constant domain models. The persistence axiom $P(\vec{t}) \rightarrow \Box P(\vec{t})$ is valid by the persistence condition on V . Standard S4 and first-order axioms are valid in all such models.

Completeness: Suppose $\not\models_{\text{FOS4}_P} \phi$. We construct a tree model where ϕ fails.

Step 1: Canonical Model. Construct the canonical model $\mathcal{M}^c = (W^c, R^c, D^c, V^c)$ for FOS4_P :

- W^c is the set of maximally consistent sets of formulas
- $\Gamma R^c \Delta$ iff for all $\Box\psi \in \Gamma$, we have $\psi \in \Delta$
- D^c is the set of equivalence classes of closed terms under provable equality
- $[\vec{t}] \in V^c(P, \Gamma)$ iff $P(\vec{t}) \in \Gamma$

The domain is constant because both Barcan formulas hold. Persistence of V^c follows from the persistence axiom, as in the propositional case.

Since $\not\models \phi$, there exists $\Gamma_0 \in W^c$ with $\neg\phi \in \Gamma_0$.

Step 2: Unraveling. Unravel $\mathcal{M}^c$ into a tree $\mathcal{T} = (W^\mathcal{T}, R^\mathcal{T}, D^c, V^\mathcal{T})$ rooted at Γ_0 :

- $W^\mathcal{T}$ consists of finite R^c -paths starting from Γ_0
- $(\Gamma_0, \dots, \Gamma_n) R^\mathcal{T} (\Gamma_0, \dots, \Gamma_n, \Gamma_{n+1})$, closed under reflexivity and transitivity
- $[\vec{t}] \in V^\mathcal{T}(P, (\Gamma_0, \dots, \Gamma_n))$ iff $[\vec{t}] \in V^c(P, \Gamma_n)$

The projection $p : (\Gamma_0, \dots, \Gamma_n) \mapsto \Gamma_n$ is a bounded morphism preserving satisfaction. The tree $\mathcal{T}$ has constant domain D^c and persistent valuation (inherited from $\mathcal{M}^c$).

Thus $\mathcal{T}, (\Gamma_0) \models \neg\phi$, giving us a tree counter-model. $\square$

Proposition 3.6.[Computational Realizability for Tree Models] Any tree model $\mathcal{M} = (W, R, D, V)$ with at most countable branching and at most countable domain can be realized by a computational model.

Proof: We construct a computational model $\mathcal{M}^c$ realizing $\mathcal{M}$.

Assigning measures to worlds. We assign to each world $w \in W$ a non-negligible set $S_w \subseteq \Omega^c$ such that:

- 1) $S_{w_0} = \Omega^c$ (the root gets the full sample space)

- 2) If w has children $w_1, w_2, \dots$ (at most countably many), the sets S_{w_i} partition S_w
- 3) Each S_w is non-negligible

The partition is constructed using bits of the random tape ρ . At each node w with children $w_1, \dots, w_k$ (finite case), we read $\lceil \log_2 k \rceil$ fresh bits from ρ to determine which child the trace belongs to. For countably infinite branching, we use a prefix-free encoding: child w_i gets traces whose next bits are $0^{i-1}1$ (i.e., $i-1$ zeros followed by a one). This assigns measure $1/2^i$ to S_{w_i} , with $\sum_i 1/2^i = 1$.

Crucially, determining which S_w a trace ρ belongs to is **PPT-computable**: given ρ , one simply reads the appropriate bits at each level of the tree. The number of bits read is proportional to the depth, and the tree structure is fixed (independent of η).

For internal nodes, $S_w = \bigcup \{S_u \mid u \text{ is a descendant of } w\}$.

Realizing the domain. Since D is at most countable, enumerate it as $d_1, d_2, \dots$ and assign distinct bitstrings $b_1, b_2, \dots$. Define constant algorithms α_{d_i} outputting b_i on all inputs. These are trivially PPT.

Realizing the valuation. For each predicate P and tuple $\vec{d} \in D^n$, define:

$$\llbracket P(\vec{d}) \rrbracket_{\eta, \rho} = 1 \iff \rho \in S_w \text{ for some } w \text{ with } \vec{d} \in V(P, w)$$

By persistence of V , if $\vec{d} \in V(P, w)$ and wRu , then $\vec{d} \in V(P, u)$. Since $S_u \subseteq S_w$ when wRu , the predicate holds on S_u as well.

Verification. The computational model $\mathcal{M}^C$ has as its frame the sets $\{S_w : w \in W^T\}$ ordered by reverse inclusion. This frame is isomorphic to the tree (W^T, R^T) : the map $w \mapsto S_w$ is a bijection, and $wR^T u$ (i.e., u is a descendant of w) iff $S_u \subseteq S_w$ (i.e., $S_w R^C S_u$).

By construction, the valuation also corresponds: for atomic $P(\vec{d})$,

$$\begin{aligned} \mathcal{M}, w \models P(\vec{d}) &\iff \vec{d} \in V(P, w) \iff \\ P(\vec{d}) \text{ holds throughout } S_w &\iff \mathcal{M}^C, S_w \models P(\vec{d}) \end{aligned}$$

The first equivalence is the definition of satisfaction in $\mathcal{M}$. The second follows from persistence: $\vec{d} \in V(P, w)$ implies $\vec{d} \in V(P, u)$ for all descendants u , and the traces in S_w are partitioned among descendants, so $P(\vec{d})$ holds at each trace iff it holds at the corresponding leaf, iff it holds at w by persistence. The third is the definition of atomic satisfaction in Kripke semantics.

Since the frames are isomorphic and atomic satisfaction corresponds, satisfaction of all formulas corresponds:

$$\mathcal{M}, w \models \psi \iff \mathcal{M}^{KC}, S_w \models_{S_4} \psi$$

This follows by a straightforward induction: negation, conjunction, and quantifiers are interpreted identically in both models, and for $\Box\psi$:

$$\begin{aligned} \mathcal{M}, w \models \Box\psi &\iff \forall u (wR^T u \Rightarrow \mathcal{M}, u \models \psi) \\ &\iff \forall u (S_u \subseteq S_w \Rightarrow \mathcal{M}^{KC}, S_u \models_{S_4} \psi) \text{ (by IH and frame iso.)} \\ &\iff \mathcal{M}^{KC}, S_w \models_{S_4} \Box\psi \text{ (in the tree-structured frame)} \end{aligned}$$

Note that the computational model $\mathcal{M}^C$ we construct uses the tree-structured frame $\{S_w : w \in W^T\}$ with accessibility given by reverse inclusion among these sets. This is a valid S4 frame (reflexive and transitive), and it is a sub-frame of the full poset of all non-negligible sets. The semantics of $\Box$ in this model quantifies only over the worlds in our frame (the S_u 's), not over arbitrary non-negligible subsets. $\square$